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REVIEW

Inter-Organ Communication in Homeostasis and Disease

Obesity and chronic kidney disease

Zongmiao Jiang,[1] Yao Wang,[2] Xue Zhao,[1] Haiying Cui,[1] Mingyue Han,[1] Xinhua Ren,[1] Xiaokun Gang,[1] and Guixia Wang[1]

[1]*Department of Endocrinology and Metabolism, The First Hospital of Jilin University, Changchun, China and* [2]*Department of Orthopedics, The Second Hospital Jilin University, Changchun, China*

Abstract

The prevalence of obesity has increased dramatically during the past decades, which has been a major health problem. Since 1975, the number of people with obesity worldwide has nearly tripled. An increasing number of studies find obesity as a driver of chronic kidney disease (CKD) progression, and the mechanisms are complex and include hemodynamic changes, inflammation, oxidative stress, and activation of the renin-angiotensin-aldosterone system (RAAS). Obesity-related kidney disease is characterized by glomerulomegaly, which is often accompanied by localized and segmental glomerulosclerosis lesions. In these patients, the early symptoms are atypical, with microproteinuria being the main clinical manifestation and nephrotic syndrome being rare. Weight loss and RAAS blockers have a protective effect on obesity-related CKD, but even so, a significant proportion of patients eventually progress to end-stage renal disease despite treatment. Thus, it is critical to comprehend the mechanisms underlying obesity-related CKD to create new tactics for slowing or stopping disease progression. In this review, we summarize current knowledge on the mechanisms of obesity-related kidney disease, its pathological changes, and future perspectives on its treatment.

chronic kidney disease; glomerulus; obesity; proximal tubular; weight loss

INTRODUCTION

Obesity is characterized by excessive fat mass and an expansion of adipose tissue resulting from adipocyte hyperplasia and hypertrophy, which causes weight gain (1). Moreover, obesity is a major public health concern, leading to serious health and economic burden. In 2014, the age-adjusted prevalence of obesity was 35% among men and 40.4% among women in the United States (2), and it is estimated to increase by 50% in 2030 (3). In China, the prevalence of overweight and obesity also demonstrates a rapid growth trend. Approximately 15% of children and 46% of adults are overweight and obese (4). As a result, obesity has become a major global health problem, and its prevalence is dramatically increasing worldwide.

To our knowledge, a close relationship exists between obesity and cardiovascular disease (CVD), diabetes, and several cancers (such as breast, colon, and endometrial cancer) (5). Overweight, obesity, and metabolic syndrome (MS) have recently emerged as strong independent risk factors for chronic kidney disease (CKD) and end-stage renal disease (ESRD). It has been nearly a century since the first studies between obesity and kidney disease were conducted (6). Since then, research on the relationship between obesity and kidney disease has developed. Hsu et al. (7) have identified overweight or obesity as a strong and potentially modifiable risk factor for the development of ESRD through a large cohort study (320,252 adults) in Northern California. Iseki et al. (8) reported similar findings in a Japanese cohort; that is, obesity additionally increases the relative risk of CKD for older individuals (aged > 65 yr). In short, growing evidence shows that obesity and kidney disease are closely linked.

The link between obesity and CKD is complex. Numerous factors have played important roles in the development of kidney disease in obesity, including insulin resistance (IR), lipotoxicity, adipocytokine dysregulation, hypertension, and enhanced glomerular blood pressure (9, 10). Clinically, obesity-related CKD is characterized by proteinuria, glomerulomegaly, progressive glomerulosclerosis, and reduced kidney function (11). Therefore, the prevention and treatment of obesity-related CKD are particularly important. In the past, weight loss (lifestyle therapies and bariatric surgery) and renin-angiotensin-aldosterone system (RAAS) blockage were the primary therapeutic methods in these patients (12). Thus, based on available research, the development of kidney damage caused by obesity is discussed in depth in this article, as well as medicines for treating obesity-related CKD.

Correspondence: X. Gang (gangxk@jlu.edu.cn); G. Wang (gwang168@jlu.edu.cn).
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CLINICAL FEATURES AND DIAGNOSIS OF OBESITY-RELATED CKD

Incidence and Clinical Presentation

The incidence of obesity-related CKD is increasing annually, and an increasing number of people are concerned about it. According to Wang et al. (13), obesity is associated with 24%–33% of all renal disease cases in the United States. Kambham et al. (14) found that the incidence of obesity-related glomerulopathy biopsies gradually increased from 0.2% in 1986–1990 to 2.0% in 1996–2000, a 10-fold increase in 15 yr. Recently, Hu et al.(15) reported an analysis of 34,630 primary kidney biopsy cases at Zhengzhou University in China and showed that the annual incidence of obesity-related glomerulopathy increased from 0.86% in 2009 to 1.65% in 2018.

The incidence of obesity-related CKD is increasing every year. Clinically, it is characterized as a proteinuric renal illness in patients with obesity who do not have clinical or histological evidence of other renal disorders and generally starts slowly, with microalbuminuria or clinically significant proteinuria as the major symptoms, with or without renal impairment, and a few individuals have microscopic hematuria or nephrotic syndrome (16). Other clinical complications include hypertension (17), hyperlipidemia (18), and obstructive sleep apnea (OSA) (19). Nicholl et al. (19) reported sleep apnea in different categories of patients with CKD. They divided 254 patients in the department of nephropathy into three groups based on their renal function on hemodialysis and found a considerable increase in the prevalence of sleep apnea (mostly obstructive), which was linked to worsening renal function, and nearly 50% of patients with CKD and ESRD experienced nocturnal hypoxia. Moreover, some studies have also found a gradient relationship between the risk of CKD and number of MS components (20, 21). Davoud et al. (22) looked into the prevalence of CKD in 6,492 individuals and found that the prevalence of CKD was markedly higher in those with MS, which increased progressively with the number of MS components and patient age. Although more clinical studies are needed to explore the clinical features of obesity-related CKD, baseline age, serum creatinine, proteinuria, and time-averaged proteinuria during follow-up have been described as risk factors associated with CKD progression (23).

Predisposing Factors

Numerous studies have found that low birth weight, low nephron number, preterm delivery, and intrauterine growth restriction are all related to an increased risk of hypertension, obesity, albuminuria, and renal disease later in life (24–26). The critical reduction in the number of functioning nephrons (such as unilateral renal agenesis and surgical removal of the renal parenchyma) is a central pathogenic factor in hyperfiltering nephropathies. In the postoperative follow-up of patients who underwent unilateral nephrectomy, it was found that patients with a higher body mass index (BMI) were more likely to have proteinuria and impaired renal function (27). In addition, obesity may play a synergistic effect on the development of renal problems in patients with impaired nephron function. A retrospective cohort study for the diagnosis of kidney disease in children indicated that obesity and prematurity both significantly increased the risk and progression rate of kidney disease (28). It is clear that a number of factors contribute to the development and progression of obesity nephropathy. In recent years, some studies have focused on the role of genetics in obesity-related kidney disease. An elevation in the expression of genes is reported to be related to inflammatory factors, insulin resistance, and lipid metabolism in patients with obese nephropathy (29). In future, genetic and metabolomic studies of obese nephropathy patients are warranted to provide possibilities for early diagnosis and treatment.

Diagnosis

Currently, the diagnosis of obesity-related kidney disease should meet the following criteria (14, 23, 30): *1*) body mass index [BMI] $\geq$ 30 kg/m^2 (should be $>$ 28 kg/m^2 for the Chinese population) and waist circumference $>$90 cm for men and $>$85 cm for women; *2*) proteinuria at various clinical levels ($>$0.3 g/24 h), severe proteinuria is rare, hypoproteinemia and edema are rarely present, and renal function is normal or mildly abnormal; *3*) light microscopy showed a significant increase in glomerular volume with or without focal segmental glomerular sclerosis (FSGS), and electron microscopy shows fusion of epithelial cell peduncles; and *4*) other primary or secondary glomerular diseases are excluded (such as IgA nephropathy and diabetic nephropathy).

Renal plasma flow (RPF) and glomerular filtration rate (GFR) are often considered to be the classical indicators for assessing the level of renal function. Since 1957, more than 70 equations based on creatinine and/or cystatin C levels have been developed to estimate GFR (eGFR) (31). However, there is some controversy as to whether these equations accurately reflect renal function in overweight and obese people (32). In a study of patients with overweight/obesity with or without CKD, 56 creatinine/cystatin C-base equations were evaluated. Compared with measured GFR (plasma clearance of iodixol), errors in eGFR were wide for any equation and greater for equations that included weight or height (32). Therefore, some authors recommend the use of gold standard techniques (such as inulin or iohexol plasma clearance) for patients with obesity, even though these methods are rarely employed because of their scarcity and impracticality in daily clinical procedures. Although proteinuria determination is extensively used for noninvasive assessment of kidney disease, it is not necessarily an early predictor of kidney injury. To our knowledge, many metabolic changes precede the appearance of renal pathology (33). Therefore, in recent years, efforts have been made to find new biomarkers to diagnose kidney injuries, such as urinary kidney injury molecule-1, urinary cystatin C, urinary *N*-acetyl-β-D-aminoglucosidase, and urinary neutrophil gelatinase-associated lipid deposition protein (34–36). Proteomic and metabolomic studies are also useful tools for predicting kidney damage; proteomic changes can precede the development of clinically relevant diseases. Many urinary proteins and peptides are considered valuable biomarkers in the management of various diseases (including CKD). Several studies have investigated the use of urinary proteomic changes in the early detection of CKD.

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Table 1. *Pathological changes in obesity-related chronic kidney disease*

Types of Kidney Injury	Pathological Damage in Kidney	References
Glomerular injury	Glomerulomegaly	(14, 47)
	Global or focal segmental glomerulosclerosis	(14, 48)
	Low glomerular density	(49, 50)
	Podocyte injury	(51, 52)
	Glomerulosclerosis	(53)
	Mesangial expansion	(54)
Kidney tubules injury	Tubulointerstitial fibrosis	(55)
	Tubular atrophy	(56, 57)
	Proximal tubular epithelial hypertrophy	(58)
	Tubular cell apoptosis	(73, 74)

For example, CKD273 classifier, which is a panel of 273 urinary peptides, has been validated in cross-sectional and longitudinal studies, and it allows the early detection of CKD and predicts the disease course (37, 38). Translating histological studies into disease-relevant biomarkers for use in clinical practice is a complex, time-consuming, and expensive process, and there is still a long way to go before they can be truly used to diagnose disease.

As a new biomarker, ectopic lipid buildup in the kidney (fatty kidney) offers potential. Renal ultrasonography (US) and ultrasound elastography, computed tomography (CT), and magnetic resonance imaging (MRI) are the most often utilized imaging modalities for evaluating fatty kidneys (39). US is a noninvasive, simple, and safe imaging test that can be performed as many times as needed. As such, it is the primary imaging modality for assessing the kidneys (40). It is used to diagnose kidney disease by observing changes in the kidney size, morphological structure, parenchymal thickness, and echogenic intensity, whereas ultrasound-guided percutaneous renal aspiration biopsy is of great value in the diagnosis of CKD (41). Similarly, CT allows for a precise noninvasive high-quality examination of kidney structure and function and assessment of ectopic adipose tissue depots associated with obesity-related CKD by quantifying adipose tissue density in Hounsfield units (42). Moreover, MRI provides a clearer view of the kidney, differentiates the renal cortex from the renal medulla without contrast, and does not involve ionizing radiation (43, 44).

PATHOPHYSIOLOGY OF OBESITY-RELATED CKD

As a result of the obesity pandemic, the number of kidney biopsies performed on patients with obesity has increased. In 1974, Weisinger et al. (45) were the first to report that patients with severe obesity could have large amounts of proteinuria with pathological biopsies suggesting the presence of kidney damage. Glomerulomegaly with mesangial proliferation, matrix buildup, and decreased density of more focally effaced and hypertrophied podocytes with milder foot process fusion are all common findings in addition to the sclerosed glomeruli (Table 1) (Fig. 1) (46). Clinically, it can be differentiated from the idiopathic structure by its

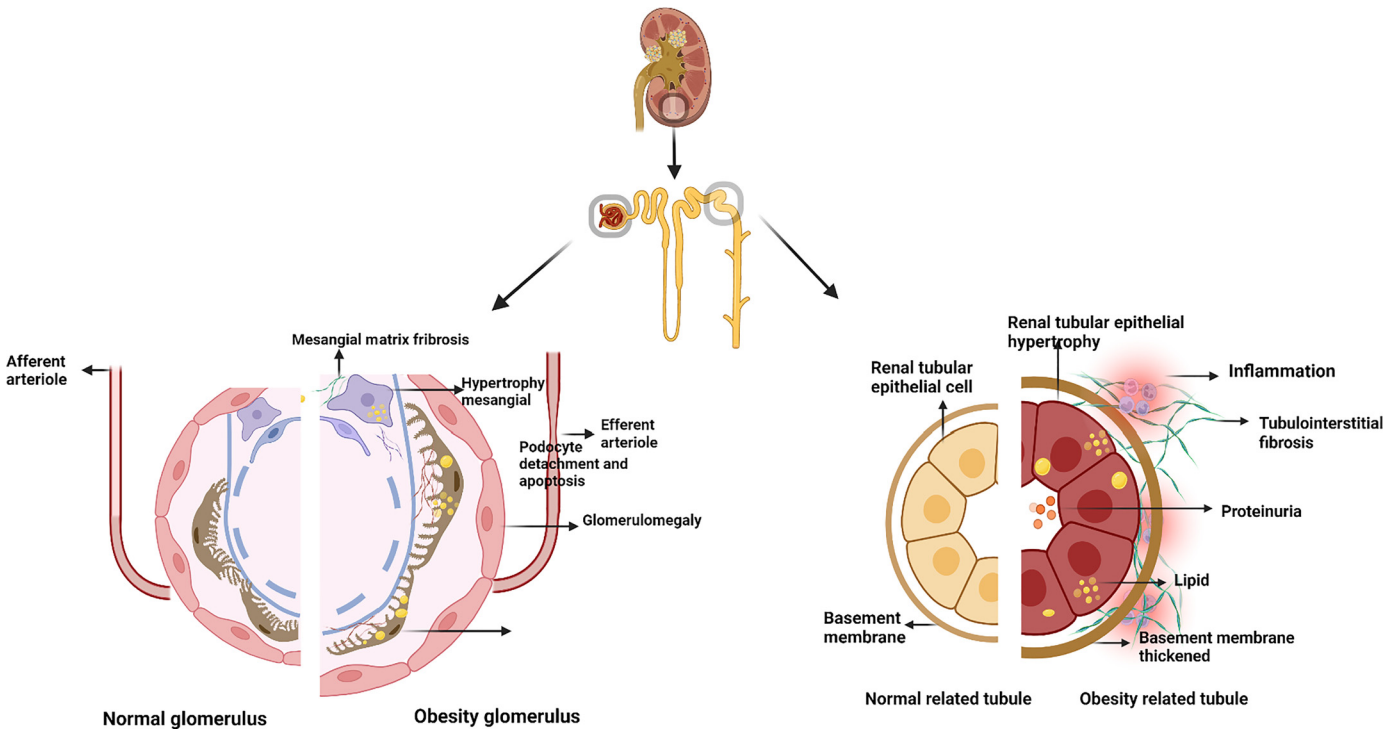

Figure 1. Pathophysiological changes in obesity-related chronic kidney disease. Obesity-associated glomerular changes include glomerulomegaly and low glomerular density, which lead to the proliferation of mesangial cells, mesangial matrix, and fibrotic lesions of the mesangial matrix, along with the compensatory hypertrophy of podocytes, loss of the podocyte foot process, and podocyte failure. Finally, FSGS or glomerulosclerosis leads to severe renal damage. Obesity-associated tubular lesions are characterized by thickening of the tubular basement membrane, tubular epithelial cell hypertrophy, apoptosis, tubular interstitial fibrosis, and inflammatory changes. Created with BioRender.com. FSGS, focal segmental glomerular sclerosis.

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slower progression of proteinuria and renal insufficiency and lower incidence of nephrotic syndrome.

Glomerulomegaly and FSGS together constitute obesity-related kidney disease. In the Columbia series, age- and sex-matched normal control samples had a mean glomerular width of 169 μm, compared with 226 μm in biopsy samples from patients with obesity-related kidney disease (14). Subsequently, foot cell damage is exacerbated by the deleterious consequences of increased intraglomerular pressure. A unique and diagnostically helpful feature of obesity kidney disease, compared with primary FSGS, is the slight disappearance of the foot processes. In a comparative study, the mean rate of foot processes loss in obesity kidney disease was 40%, compared with 75% in primary FSGS (14), and immunofluorescence shows nonspecific deposition of IgM and C3 in sclerotic and hyaline degenerative lesions without immune complex deposition. In addition, mesangial cells and tubular epithelial cells' cytoplasms may develop focal lipid vacuoles, particularly in the proximal tubule, due to excessive lipid accumulation (10).

Obesity-Related Glomerulopathy

The diameters of all glomerulus samples or those sectioned via the hilus, which is in the middle region of the glomerular globe, are measured to determine glomerulomegaly (47, 50). Through other means, the serial portions of a glomerulus are employed to measure the glomerular volume. Glomerulomegaly is associated with mesangial proliferation, matrix buildup, and hypertrophied podocytes, as well as milder foot process fusion (10, 47, 54). Humans are born with a fixed number of nephrons, which must grow, and their single-nephron GFR increases to satisfy the metabolic needs of obesity (59, 60).

Two concepts have been proposed to explain obesity-related glomerular abnormalities. First, due to altered hemodynamic in people with obesity, the glomerular inlet arteries are dilated, which in turn triggers glomerular hyperfiltration (GH) (61) and this induces basement membrane expansion and glomerulomegaly. Second, there is increased water and salt reabsorption by the proximal tubule, which in turn leads to a deactivation of the tubuloglomerular feedback. In the Fischer rat model, increase in proteinuria and glomerular volume was linearly related to increasing body weight (62).

Podocytes

Podocytes are terminally differentiated epithelial cells that extend to wrap around the basement membrane of glomerular capillaries in the glomerulus (63, 64), a key component of the glomerular filtration barrier. With the patient gaining excessive weight, increasing single-nephron GFR and consequently increased podocyte shear stress, raise the risk of developing podocytopathy (52). To compensate for the reduced filtration area of the renal unit, the remaining nephrons grow in size (hypertrophy), which might lead to maladaptive podocyte hypertrophy. Hypertrophic podocytes may also be unable to maintain a normal foot process structure, resulting in increased local shear stress and subsequent podocyte separation (65, 66). In one study, individuals with obesity-related glomerulopathy had lower podocyte density (134 ± 54.1 vs. 245 ± 48.4) and number (615 ± 235 vs. 714 ± 103), and changes in podocytes were linked to the severity of proteinuria and renal function impairment in these patients (51). Furthermore, one study suggested that diet-induced obesity caused increased renal lipid accumulation and glomerulosclerosis in C57BL/6J mice (67).

FSGS

FSGS is defined as segmental consolidation of the glomerular tuft by extracellular matrix and/or hyaline that causes capillary obliteration (68). FSGS is a complex disease that develops following podocyte injury caused by various factors, some of which are known and others unknown. D'Agati et al. (69) initially proposed dividing FSGS into primary (idiopathic) and secondary forms. In some cases of primary FSGS, autoimmune-mediated podocyte destruction could be a risk factor. By contrast, secondary FSGS is discovered after a substantial reduction in renal mass, when the workload of the remaining nephrons is increased (70). Of the five histological subtypes of FSGS (not otherwise specified, periportal, cytosolic, apical, and collapsed), the periportal type is more common in obesity-associated nephropathy, and it may be associated with the increased diameter of the afferent arteriole and reduced podocyte density. Obesity-induced GH per surviving nephron has been proven in human research; moreover, it negatively affects long-term kidney function in patients with loss of renal mass and subsequent FSGS (48, 71, 72).

Proximal Renal Tubule

In patients with obesity, various factors such as hypertension, hyperfiltration, and lipotoxicity can cause tubular atrophy (56, 57), tubular cell apoptosis (73, 74), tubulointerstitial fibrosis (55), proximal tubular epithelial hypertrophy, among other lesions. Obesity and overweight are associated with increased GFR, RPF, and/or filtration fraction. A diabetes study showed that glomerular expansion is associated with tubule hypertrophy (75). Chronic obesity, elevated arterial pressure, renal vasodilation, GH, neurohumoral activation, and metabolic changes may lead to glomerular damage and further damage from renal pressure natriuresis, leading to more severe hypertension and progressive loss of renal function (76). Meanwhile, increased intrarenal pressure and abdominal pressure due to perirenal fat (PF) may impair stress natriuresis and lead to obesity-associated hypertension (77). As with other forms of hypertension, the rise in blood pressure associated with obesity is accompanied by a reduction in urinary sodium. Experimental evidence in animal models revealed that proximal tubular hypertrophy and enhanced proximal tubular sodium reabsorption play a key role in the pathophysiology of GH in diabetic kidneys (78). Clinical overabundant lipid accumulation is most common in lipoid nephrosis, a condition in which enormous oval fat bodies form in the proximal tubular cells in the case of nephrotic syndrome. The excess nonesterified fatty acid burden in proteinuria finally causes tubulointerstitial damage, with a strong role in decreased autophagy response, according to experimental investigations (79). Recently, similar findings have been made in studies of obesity-related kidney disease. Tobar et al. (58) showed that in persons with proteinuria, obesity-related GH is associated with proximal tubular

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epithelial hypertrophy. Excessive weight gain increases tubular reabsorption and impairs stress natriuresis in part by activating the sympathetic and renin-angiotensin systems and physically compressing the kidneys. With prolonged obesity, the kidneys also undergo structural changes that eventually lead to the loss of kidney unit function (80).

In summary, glomerulomegaly with FSGS is a typical renal histopathological finding in obesity-related CKD (81). However, these findings are not CKD-specific; therefore, it is crucial to exclude diseases closely associated with obesity, such as hypertensive nephrosclerosis or diabetic glomerulosclerosis, when diagnosing this entity (47). These findings highlight the usefulness of kidney biopsy as a technique to guide the treatment and prognosis of patients with severe obesity and kidney disease.

PATHOGENESIS OF OBESITY-RELATED CKD

As obese individuals often present with several components of metabolic syndrome (such as hypertension, insulin resistance, and dyslipidemia), it is difficult to clearly define whether the effect of obesity on CKD is independent of these metabolic disturbances. Currently, there is growing evidence that metabolically healthy obesity (MHO) is associated with a higher risk of CKD events (82). A population-cohort study from South Korea showed that individuals with stable MHO had a higher risk of CKD than stable metabolically healthy, nonobese individuals, supporting the role of obesity in CKD development (83). Another study of patients with MHO noted the development of other components of metabolic syndrome over time, which later led to kidney damage (84). The possible mechanisms of obesity leading to CKD include inflammation, hemodynamic abnormalities, metabolic disorders, and hormone synthesis such as angiotensin-II (ANG II) (85, 86).

Association of Inflammation with Obesity-Related CKD

Recent evidence shows that kidney inflammation is very important in initiating the development and progression of obesity-related kidney disease. Several investigations have demonstrated that adipose tissue, particularly visceral adipose tissue, is a primary source of cytokine release in MS (87, 88), such as leptin, adiponectin, tumor necrosis factor (TNF), monocyte chemoattractant protein-1, transforming growth factor, and ANG II (89). Persistent inflammation is also an essential feature of CKD, which contributes to progressive kidney fibrosis, endothelial dysfunction, and activation of glomerular and tubular epithelial cells, which results in the release of inflammatory chemicals, which attract immune cells to the damaged kidney (90).

The NF-κB pathway, reactive oxygen species, and activation of protein kinases, such as the mitogen-activated protein kinase cascade and the JAK-STAT signaling pathway, are key inflammatory processes involved in renal injury. TNF-α is one of the most important mediators of adipose tissue inflammation and IR development among proinflammatory adipokines (91, 92). Increased TNF-α levels are frequently linked to increased production of monocyte chemotactic protein-1 (MCP-1), a chemokine produced by adipocytes and macrophages that have been linked to increased fat accumulation (93, 94). The levels of interleukin (IL)-6, TNF-α, IL-1, and MCP-1 in kidneys from a diet-induced obesity rat model are higher, which is related to increased kidney fibrosis and sclerotic lesions (95). Moreover, TNF-α deficiency protects the kidney from obesity-induced albuminuria and kidney damage (96).

Altered Renal Hemodynamics in Obesity

Conventionally, GFR is determined by RPF, hydraulic pressure gradient across the kidney membrane, and ultrafiltration coefficient. According to the National Kidney Foundation, normal results range from 90 to 120 mL/min/1.73 m^2. Older people will have lower than normal GFR levels because GFR decreases with age (97).

GH is thought to represent a relationship between obesity and renal damage (98). Hyperinsulinemia, hypertension, upregulated RAAS, and increased sodium/glucose absorption in the proximal tubules (affecting the tubular-glomerular feedback) may all contribute to GH by changing afferent to efferent arteriolar tone in metabolically unhealthy obesity (53). In a prospective cohort analysis of 158,365 participants, Reynolds et al. (99) investigated the association between blood pressure and the incidence of ESRD, and they found that participants with *stage 2* hypertension had a 3.4-fold higher chance of acquiring glomerulonephritis-related ESRD and a 2.2-fold higher risk of developing nonglomerulonephritis-related ESRD than those with normal blood pressure. Renal hypertrophy is the structural counterpart of renal hyperfiltration, characterized by an increase in the glomerular tuft volume, Bowman's gap, tubular epithelium, and tubular lumen at the nephron level (14). In a study of nondiabetic African Americans, obesity and overweight are linked to higher GFR, effective renal plasma flow (ERPF), filtration fraction, and high prevalence of GH (59). GH increased proximal tubular flow and reabsorption of small- and high-molecular-weight solutes, resulting in glomerular hypertrophy, tubulointerstitial inflammation, and fibrosis; meanwhile, GH-associated physical forces exert mechanical stress on the filtration barrier and postfiltration structure, which eventually leads to podocyte loss and FSGS. Therefore, in obesity-induced proteinuric CKD, reducing hyperfiltration is an important treatment target.

Lipotoxicity and CKD

Lipids are a significant component of the normal kidney, accounting for ∼3% of its wet weight (100). In renal disease, fat infiltration was initially seen more than a century ago (101). Building on this, Moorhead in his study (102) hypothesized a critical role for lipid toxicity in the pathogenesis of CKD over 30 years ago. The exact mechanisms whereby obesity caused CKD remain unclear. Obesity-related kidney disease is characterized by a complex and integrated set of processes. The adipose tissue deposited around the kidney exerts local pressure on the kidney and affects renal hemodynamics. On the contrary, the adipose tissue influences the kidneys through various secretory agents, including cytokines, adipokines, and metabolites, all of which are necessary for optimal renal function. This section discusses in detail the association between lipotoxicity and nephropathy.

Although dyslipidemia is thought to be a consequence of kidney disease, a large number of clinical studies supported

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that altered lipid metabolism may contribute to the pathogenesis and progression of kidney disease. A Mendelian randomized study of a Chinese population noted higher blood TG levels could be causally associated with a higher risk of CKD (103). Moreover, a cohort study in the UK population showed that elevated TG and reduced HDL-C levels were independently associated with the onset of advanced CKD (104). Interestingly, conclusions on the relationship between serum lipids and the development of CKD are divergent and conflicting. Lipid and CKD cohort data from another Mendelian randomization analysis support the causal relationship between genetically greater HDL cholesterol concentration and better kidney function, however, no association between genetically altered LDL cholesterol or triglyceride concentrations and renal function was observed (105). This may have been caused by differences in the ethnicities of the groups studied. More basic research may be needed in the future to confirm the mechanisms by which lipids affect kidney pathology.

PF, the adipose tissue that surrounds the kidneys, was formerly assumed to provide mainly mechanical kidney support; nevertheless, investigations have shown that the PF has a stronger association with kidney illnesses than other visceral fat deposits in obesity or metabolic disorders (106). PF tissue is a metabolically active adipose tissue depot, and it plays endocrine and paracrine roles in glucose and lipid homeostasis and inflammation by generating and secreting adipokines (107). Increasing evidence suggests that PF thickness is associated with CKD risk and may be used as a predictor of reduced GFR and increased incidence of proteinuria in individuals who are obese/overweight (108, 109). The detailed mechanisms by which PF triggers and exacerbates chronic kidney injury have not been fully elucidated, but several pathways are postulated. PF accumulation may directly compress the renal vasculature and parenchyma, leading to increased interstitial hydrostatic pressure, stimulation of renin release, glomerular filtration, and sodium tubular reabsorption, all of which accelerate the progression of renal disease and ultimately lead to decreased GFR (110). On the contrary, excess PF can trigger kidney injury through the paracrine or systemic secretion of adipokines and inflammatory factors and activation of the sympathetic and RAAS systems. For example, leptin, adiponectin, TNF-α, and IL-6 can lead to elevated GFR and increased urinary albumin excretion by altering renal hemodynamics and damaging the vascular endothelium (111, 112).

Adiponectin.

Adiponectin is a protein that is mainly secreted by adipocytes and expressed in hepatic parenchymal cells, myocytes, and epithelial cells (113–115). As a polypotent hormone, it has anti-inflammatory, antifibrotic, insulin-sensitizing, and cardioprotective effects (116, 117). Reduced adiponectin levels are associated with MS (118), obesity, type 2 diabetes mellitus (T2DM) (119), and hyperlipidemia. Adiponectin levels appear to be directly linked to renal function, and it is found mostly in the artery endothelium, smooth muscle cells of the kidneys, and capillary endothelium (120). Ohashi et al. (121) performed subtotal (5/6) nephrectomy on adiponectin knockout and wild-type mice. They noted significantly worse urine albumin excretion, glomerular hypertrophy and tubulointerstitial fibrosis in the adiponectin knockout mice compared with wild-type mice. Moreover, in patients with obesity, Lakkis et al. (122) discovered an independent inverse connection between adiponectin and albuminuria in people who were nondiabetic overweight and obese. In a study in Japan, Yano et al. investigated the relationship between adiponectin and low-grade albuminuria in individuals who were nondiabetic obese and lean. They discovered that patients with obesity having low adiponectin levels excreted considerably more urine albumin than patients with obesity having high adiponectin levels. This remained true even after correcting for age, sex, systolic blood pressure, and fasting glucose level (123).

Leptin.

Leptin, coded for by the Lep^{ob} gene (124), is an adipocyte-derived hormone synthesized mainly in white adipose tissue (125). As a tiny peptide, leptin is mostly eliminated by the kidney via megalin-mediated tubular absorption, endocytosis, and intracellular breakdown (126, 127). Patients with CKD have increased leptin levels due to reduced GFR; moreover, leptin induces oxidative stress (128), endothelial dysfunction, and enhances platelet aggregation (129), which play important roles in CKD development and progression. Studies have found significantly higher serum leptin levels in the obese population than in the normal-weight ones (130, 131), and hyperleptinemia may contribute to the development of glomerulosclerosis and profibrotic effects on mesangial cells (132). Meanwhile, leptin levels continue to increase as renal function gradually declines, which may eventually induce cachexia. Kokot et al. (133) showed that the sharp increase in leptin with BMI in patients with renal failure compared with those having a normal renal function is due to concomitant inflammation, hyperinsulinemia, and low erythropoietin levels in patients with CKD.

RAAS and CKD

Angiotensinogen (AGT), the obligate precursor of all bioactive angiotensin peptides, is sequentially cleaved to ANG I and ANG II by rennase and angiotensin-converting enzyme (ACE), respectively (134). AGT is localized in the proximal tubules of the kidney, whereas ACE is present in large numbers in the proximal and distal tubules and collecting ducts (135). Obesity is associated with increased RAAS activity, particularly in visceral adipose tissue, as seen by the higher levels of AGT, renin, ACE, and aldosterone in patients with obesity and animals in multiple investigations (136). RAAS and sympathetic nervous system activation cause impaired natriuresis and increased renal tubular salt reabsorption, resulting in volume expansion and hypertension (137). Furthermore, aldosterone is a steroid hormone that increases the ability of the kidney to reabsorb ions and water molecules. Aldosterone boosts the effects of ANG II, causes vascular inflammation and remodeling, and stimulates mineralocorticoid receptors in the kidney. It also can lead to kidney inflammation, fibrosis, podocyte damage, and proliferation of thylakoid cells (138). Several studies have found that inhibitors or blockers of RAAS can potentially reduce obesity-associated hypertension and thus improve renal dysfunction (139, 140).

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IR and CKD

Obesity and IR are key risk factors for a reduction in GFR in the kidney and the initiation and progression of CKD (141, 142). Insulin can affect practically all renal cell types, including mesangial cells, podocytes, and tubular epithelial cells (143). The presence of both obesity and IR contributes to the inappropriate activation of the sympathetic nervous system (SNS) and RAAS, as well as the involvement of adipokines, oxidative stress, and inflammatory pathways, all of which are involved in kidney dysfunction (144). These negative effects are mediated by decreased endothelial nitric oxide production, increased expression of the mesangial angiotensin II type 1 receptor, and increased synthesis of endothelin-1 and growth factors (145). Several studies have found a correlation between IR and decreased renal function (146–148). In a large population-based cross-sectional study of an American Indian population, IR was linked to sex, residential location, smoking, and a high-risk cardiometabolic profile in a cross-sectional study; in this population, IR was related to the development of CVD and impaired kidney function (149).

Dysbiosis of Gut Microbiota and CKD

The gut microbiota has recently been identified to play a key role in the development of metabolic illnesses, and it is thought to be an endocrine organ involved in the regulation of energy balance and host immunity (150). The human gut microbiota consists of up to 100 trillion microorganisms that are largely symbiotic with their human hosts (151). The gut microbiota plays an important role in physiological and disease states, including obesity, diabetes, CKD, CVD, and cancer (152, 153). Previous data have shown that the rates of obesity-associated thick-walled phylum of Bacteroides and Bacteroides were higher in children with obesity than in those with normal weight (154). Moreover, altered quantitative and quantitative gut microbiota was found in patients with CKD and ESRD (155), and preliminary evidence suggests that toxic products produced by dysbiosis of the gut flora may contribute to the progression of CKD and CKD-related complications (156). Intestinal dysbiosis could be caused by iatrogenic factors or by uremia itself (157). When renal function is lost, urea is secreted into the gastrointestinal system. The subsequent breakdown of urea by urease, which is expressed by some gut microbes, results in the creation of significant amounts of ammonia, which may inhibit the proliferation of commensal bacteria (158). Uremic individuals have significantly higher counts of both aerobic and anaerobic organisms in the duodenum and jejunum, which are generally not densely colonized by bacteria in healthy people (159).

Mounting evidence revealed that altered gut microbiota, leaky gut syndrome, and endotoxemia have a role in initiating or perpetuating the inflammatory state (160, 161). As stated earlier, chronic inflammation plays a complex role in obesity and CKD. The gut microbes break down tryptophan to indole, which is then converted to indole-3-acetic acid in the intestine and indoxyl sulfate (IS) in the liver (162); both toxins are involved in atherogenesis, vasodilation, and oxidative stress (163). A meta-analysis of 10 controlled feeding trials involving 292 patients with CKD showed that increased dietary fiber intake significantly reduced indophenol sulfate, p-cresol sulfate, blood urea nitrogen, and uric acid levels in patients with CKD, with more pronounced effects in dialysis and nondiabetic patients (164).

In summary, adipose tissue influences the kidneys through various secretory agents, including cytokines, adipokines, and metabolites, all of which are necessary for optimal renal function. Therefore, excessive ectopic accumulation of fat induces kidney damage through the activation of RAAS, SNS, inflammatory factors, oxidative stress, etc. (Fig. 2). In addition, some studies investigating gene expression in the glomeruli of patients with CKD have demonstrated that gene polymorphisms are strongly associated with CKD (165), but further research applying genetic testing is necessary for the diagnoses of and guidance in nephropathy treatment.

TREATMENT OF OBESITY-RELATED CKD

Early kidney injury caused by obesity is mainly manifested by glomerular hyperfiltration and hyperperfusion when dietary and exercise interventions to reduce weight can reverse kidney injury (166, 167). When kidney injury gradually develops in the later stages, if severe proteinuria and renal insufficiency appear, it will be more difficult to treat. The final goal is to delay ESRD development and reduce CKD-associated mortality. The protection of kidney damage should combine weight reduction with the proper control of associated risk factors, hypertension, MS, diabetes, and dyslipidemia (168). Therefore, therapeutic strategies,

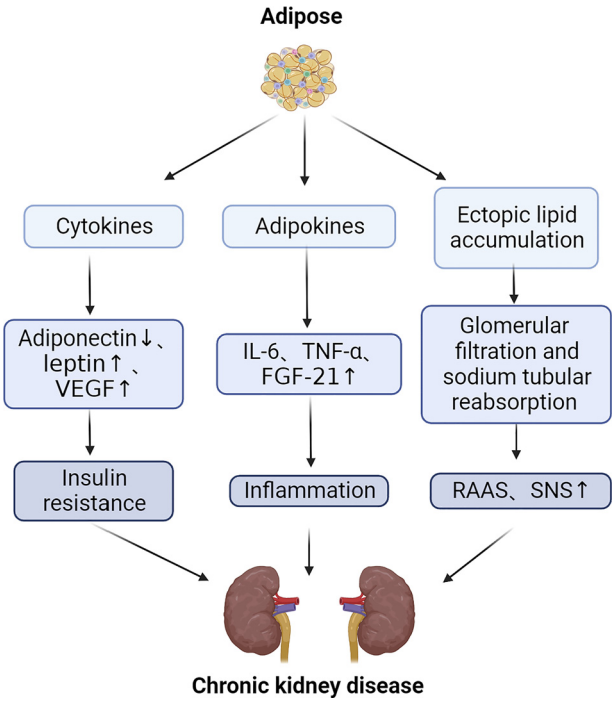

Figure 2. Influence of obesity on the progression of chronic kidney disease. A chronic high-fat diet can lead to the development of obesity when large amounts of fat accumulate in the kidney. It can accelerate kidney damage through inflammation, lipotoxicity, and IR. Prolonged fat accumulation can further lead to glomerular and tubular disease. Created with BioRender.com. IR, insulin resistance.

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which combine weight loss, lifestyle modifications, and pharmacological therapies, may help restore body weight to a more favorable level and prevent or slow down CKD progression (Table 2).

Weight Loss

Weight loss is likely to reverse proteinuria by reducing multiple toxicities, such as through better control of blood pressure, hyperglycemia, and dyslipidemia, improvement in insulin sensitivity, decrease in circulating leptin levels, reversal of GH, and reduction in RAAS activation.

Diet and exercise intervention.

Some studies have reported that weight loss slows the progression of renal damage, which shows a significant reduction in the urine albumin excretion rate and improves the eGFR (182, 183). Brucl et al. (184) reported that a very low-calorie ketogenic diet is a safe and effective dietary intervention in patients with obesity and mild CKD, and 27.7% of patients with mild renal failure reported normalization of GFR after dietary intervention. Aerobic exercise and resistance training have been recommended to have antiobesity effect and influence inflammation, endothelial dysfunction, oxidative stress, IR, and CKD progression (185). Through a 4-mo clinical observation of dietary exercise restriction and aerobic exercise in patients with severe CKD, Ikizler et al. (186) found that dietary caloric restriction mainly reduced body weight and absolute fat mass, whereas aerobic exercise largely reduced oxidative stress and inflammatory response and improved metabolic health in patients with moderate-to-severe CKD. Diet and exercise interventions have positive effects on the long-term prognosis and quality of life of patients with CKD and are an important nonpharmacological treatment modality.

Bariatric surgery.

According to the International Federation for Surgery of Obesity Global Registry data, the number of bariatric procedures reached 394,431 in 2018 (187). Some studies have compared kidney function outcomes in patients with obesity who undergo bariatric surgery to matched nonsurgery patients with obesity (137, 188, 189). Overall, these studies have consistently shown that bariatric surgery is associated with eGFR decline and a lower risk of kidney failure. In a retrospective cohort study of patients with CKD stage 3 or higher who received bariatric surgery at the KPSC health system, bariatric surgery slowed eGFR decline >50% up to 3 yr postsurgery compared with that 2 yr before surgery (137). Similar results have been reported by the Swedish Obesity Study (an ongoing, nonrandomized, matched, prospective intervention study), which shows that the long-term incidence of ESRD was reduced by >70% in the surgery group compared with the usual care control group (190). Studies have shown that weight loss surgery can reduce the risk of kidney disease to some extent (188, 191). However, we do not know yet the best bariatric procedure to protect the kidneys, which requires more clinical experience.

Antiobesity drugs.

The Federal Drug Administration (FDA) has approved several drugs for weight loss, such as liraglutide, orlistat, and Phentermine. The safety and efficacy of several glucagon-like peptide 1 receptor agonist (GLP-1 RAs) in patients with T2DM and CKD have been established in several trials, even though these studies were not designed as weight loss trials (192–194). At present, only liraglutide is FDA approved at a higher dose for weight-loss indication. The European Renal Association-European Dialysis and Transplant Association published a consensus and advocated the use of GLP-1 RAs in the treatment of patients with T2DM and CKD. GLP-1RAs can reduce cardiovascular events, all-cause mortality, and renal disease progression in patients with T2DM (195). Cook et al. (196) found that combining diet, exercise, and orlistat could significantly reduce weight and improve physical functioning in patients with obesity-related CKD, potentially allowing them the opportunity for kidney transplantation. Another prospective study of 60 patients with obesity showed that 180 days of orlistat 120 mg/3 times/day without hypocaloric diet significantly reduced BMI in all participants and albuminuria only in patients without DM (197). The potential role of different drug classes in reducing proteinuria and protecting GFR decline is far from being established; meanwhile, further work is required to determine whether these beneficial effects are maintained in the longer term.

RAAS Blockade

In obesity, the RAAS is an important system that is regulated by excess visceral adipose tissue and excess circulating insulin (198). RAAS blockade is valid in retarding renal disease progression in laboratory animals (199). Ohmura et al. (200) found that telmisartan has a definite renoprotective effect against renal injury in obese Zucker rats. Similarly, telmisartan confers an underlying protective effect on visceral adipose tissue and kidney. They compared Wistar rats fed a high-fat diet (MetS group) with rats fed high-fat diet and telmisartan (MetS-Tel); consequently, the MetS-Tel group had significantly decreased 24-h urinary protein (104.91 ± 18.25 vs. 300.88 ± 13.75 mg/24 h; $P < 0.05$) relative to the MetS group (201). However, in humans, the protective effect of RAAS blockade on the kidneys is controversial. Several clinical trials have shown that ACE inhibitors and angiotensin-receptor blockers reduced renal risk (202–204). Based on 42 randomized controlled trials (RCTS) with 14,576 patients, Feng et al. (205) reported that a combination of RAAS inhibitors was superior to monotherapy in reducing urine protein; however, such superiority does not translate into delayed ESRD progression and cardiovascular mortality. No particular guidelines have provided specific recommendations for patients with obesity-related kidney disease; thus, more clinical data are needed to support the protective effects of RAAS blockade on these patients (Fig. 3).

Fecal Microbiota Transplantation

Imbalanced gut microbiota is related to the development of obesity, and it helps convert ingested complex nutrients to short-chain fatty acids and modulate energy homeostasis (206). Fecal microbiota transplantation (FMT) is a method of changing the recipient's gut microbiota to normalize its composition and gain a therapeutic benefit (207). The efficacy of FMT in the treatment of obesity is still controversial. Bäckhed found that the transplantation of gut microbiota

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Table 2. *Treatment of obesity-related chronic kidney disease*

Author/Year, Reference	Therapy	Study Design/Study Population	Intervention/Follow-Up Time	Weight Outcomes	Kidney Outcomes
Bruci/2020 (184)	Caloric restriction (very low-calorie ketogenic diets)	Observational prospective study/patients with obesity (n = 92)	*Arm 1*: MCKD *Arm 2*: NKF/duration: 14.9 ± 8.5 wk	Both groups lost weight, but there was no statistical difference between the two groups.	27.7% of patients with MCKD reported an improvement of renal function.
Shen/2010 (167)	Food and exercise intervention	Observational prospective study/patients with ORG (n = 63)	*N*/2 yr	24 mo later, 27 patients achieved a 9.20 ± 3.78% reduction in BMI.	24 mo later, 27 patients achieved a 51.33% reduction in urine protein secretion; in patients with increased BMI, urine protein was increased by 28.78%.
Díaz-López/2021 (180)	Weight loss	Randomized controlled/overweight/obese (n = 6,719)	*Arm 1*: Mediterranean diet and lifestyle intervention *Arm 2*: usual care advice/1 yr	1-yr weight decreased 3.7 kg in intervention arm and decreased 0.7 kg in control arm.	Compared with the control group, the eGFR decline rate was about 0.6 mL/min/1.73 m^2 lower in the intervention group.
Straznicky/2011 (181)	Dietary weight loss/aerobic exercise	Randomized controlled/overweight or obese (n = 38)	*Arm 1*: WL by caloric restriction alone *Arm 2*: WLREX *Arm 3*: no treatment/12 wk	Body weight decreased by 8.2 ± 0.8% in the WL and 10.7 ± 0.9% in the WLREX groups.	Urinary albumin excretion rate decreased by 30 ± 24 in the WL and 15 ± 6 in the WLREX groups.
Liakopoulos/2020 (217)	Bariatric surgery	Observational cohort study/10,642 obese with T2MD patients	*Arm 1*: 5,321 patients with gastric bypass *Arm 2*: 5,321 patients without gastric bypass/follow-up > 4.5 yr	After 2 yr of follow-up, the average weight loss in the *arm 1* was 11.49 kg (42.03 to 30.54) and 2.31 kg (40.95 to 38.64) in the *arm 2*.	The incidence of ESRD was reduced by about 50% in patients who gastric bypass compared with the control group.
Nehus/2017 (218)	Bariatric surgery	Longitudinal Assessment cohort/242 obese teenager patients	Bariatric Surgery (Roux-en-Y gastric bypass, sleeve gastrectomy, adjustable gastric band)/3 yr	At 3-yr follow-up, BMI decreased from 50.5 kg/m^2 at baseline to 36.2 kg/m^2.	Geometric mean (95% CI) of ACR was 74 mg/g (45–121 mg/g) at baseline and decreased to 17 mg/g (10–28 mg/g) at 3 yr.
Solini/2019 (219)	Bariatric surgery	Observational prospective study/25 nondiabetic obese patients	Rouxen-Y gastric bypass/1 yr	Mean BMI and waist circumference reduction achieved nearly 30% (Excess weight loss = 71 ± 22 %).	Serum creatinine and eGFR showed no change after intervention. Bariatric surgery improves mGFR/BSA and renal and systemic vascular function.
Neff/2016 (220)	Bariatric surgery	Retrospective study/461 patients with obesity	RYGB and LAGB/5 yr	RYGB group achieved a total percentage weight loss of 24 ± 1 %, LAGB group achieved loss of 21 ± 1 %.	eGFR increased over 5 yr after RYGB (94 ± 2 to 102 ± 22 mL/min/1.73 m^2, P = 0.01) and LAGB (88 ± 1 to 93 ± 22 mL/min/1.73 m^2, P = 0.02).
Mann/2017 (221)	Antiobesity drugs	Randomized, controlled trial/9,340 T2MD.	*Arm 1*: liraglutide *Arm 2*: placebo/3.84 yr	Not reported	The prespecified composite renal outcome occurred in fewer patients in the liraglutide group than in the placebo group (268 vs. 337).
Tonneijck/2016 (222)	Antiobesity drugs	Randomized, double-blind trial/55 T2MD	*Arm 1*: sitagliptin *Arm 2*: liraglutide *Arm 3*: placebo/12 wk	Sitagliptin reduce body weight after 12 wk [1.2 kg (−2.5 to 0.05)], liraglutide reduced body weight after 6 wk [1.7 kg (−3.0 to −0.5)] and 12 wk [1.9 kg (−3.2 to −0.6)].	Neither treatment induced changes in ACR, NGAL, or KIM-1.

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Table 2.— *Continued*

Author/Year, Reference	Therapy	Study Design/Study Population	Intervention/Follow-Up Time	Weight Outcomes	Kidney Outcomes
Stenlof/2007, (223)	Antiobesity drugs	Randomized controlled trial/535 newly diagnosed type 2 diabetes naive to glucose-lowering drug treatment	*Arm 1*: topiramate 96 mg/day *Arm 2*: topiramate 192 mg/day *Arm 3*: placebo/40 wk	Weight change (coprimary) greater in topiramate arms (96 mg/day 6.6%; 192 mg/day 9.1%) vs. placebo 2.5%.	24-h urine albumin excretion greater in topiramate arms (96−16.2 mg/day; 192−15.7 mg/day) vs. placebo −1.0 mg/24 h.
Kang/2021 (224)	Antiobesity drugs	Randomized controlled trial/159 patients with overweight/obesity	*Arm 1*: exenatide *Arm 2*: glargine/24 wk	After 24 wk, in exenatied group BMI decreased 2.005 kg/m^2 at baseline, and glargine group decreased 0.060 kg/m^2.	Exenatide intervention could reduce the UAC endpoints ($P < 0.001$) whereas insulin glargine did not influence the UAC.
Morales/2015 (225)	RAAS blockade	Controlled clinical trial/71 patients with proteinuria levels >1 g/24 h who were taking spironolactone and were being treated with other RAAS blockers.	*Arm 1*: obesity group (BMI ≥ 30 kg/m^2) *Arm 2*: control group (a BMI < 30 kg/m^2)/28.9 (14–84) mo	Not reported.	Proteinuria in the obesity group decreased 59.4% (2.8 ± 2.1 vs. 1.3 ± 1.6 g/24 h, $P < 0.05$). Control group reduction 69.6% (1.9 ± 1.4 to 0.8 ± 0.5, $P < 0.05$).
Guney/2009 (226)	RAAS blockade	Prospective study/24 nondiabetic patients with CKD treated with ACEIs and/or ARBs.	*Arm 1*: 25 mg spironolactone *Arm 2*: control/6 mo	Not reported.	No significant changes in glomerular filtration rate, creatinine, and albumin, during the observation period in either group; spironolactone reduced both proteinuria and urinary TGF-β1 excretion in patients with CKD.

Data are mean ± SD or median (interquartile range), unless stated otherwise. *n* = Subjects. ACR, albumin-to-creatinine ratio; BMI, body mass index; ACEIs, angiotensin-converting enzyme inhibitors; ARBs, angiotensin II receptor blockers; BSA, body surface area; CKD, chronic kidney disease; eGFR, estimate GFR; KIM-1, kidney injury molecule 1; LAGB, laparoscopic adjustable gastric banding; MCKD, mild chronic kidney disease; mGFR, measured glomerular filtration rate; NGAL, neutrophil gelatinase-associated lipocalin; NKF, normal kidney function; ORG, obesity-related glomerulopathy; RAAS, renin-angiotensin-aldosterone system; RYGB, Roux-en-Y gastric bypass; T2DM, type 2 diabetes mellitus; UAC, urine albumin concentration; WL, weight loss; WLREX, weight loss by combined caloric restriction and aerobic exercise.

from normal obese mice to germ-free mice led to a 57% total body weight gain after 14 days (208). However, in an RCT, Allegretti randomly divided 22 patients with obesity into two groups: the group that received FMT by capsules (FMT from a single lean donor, BMI > 17.5 kg/m^2) and the placebo capsule group. After 12 wk, they did not observe significant changes in the mean BMI in either group (209). Nevertheless, to our knowledge, FMT has not been undertaken in patients with obesity-related kidney disease, but we believed that FMT might be a useful therapy (210), although it is still far from application in humans. Recently, the use of probiotics, prebiotics, and synbiotics for kidney treatment is an area of increasing interest for professionals. Prebiotics constitute an indigestible compound that is metabolized by microorganisms in the gut, thereby promoting the growth of beneficial bacteria (211). Numerous studies have shown that the administration of prebiotics, probiotics, and synbiotics in patients with CKD can reduce the levels of uremic p-cresyl, IS, and inflammatory mediators, resulting in significant improvements in endotoxemia, blood urea nitrogen levels, and quality of life (212–214). Research on the role of prebiotics and probiotics in the regulation of intestinal ecological dysregulation and the mechanisms involved in renal protection is still limited. More detailed findings from preclinical and clinical studies are needed to clarify the dose and duration of use.

Melatonin

Melatonin (MT) is an endogenous indoleamine hormone found in nearly all organisms, from simple photosynthetic bacteria to humans (215). In vertebrates, MT is produced mostly by the pineal gland, but it can be also produced by the retina, gastrointestinal tract, spleen, kidneys, and innate immune system (169). MT is a pleiotropic hormone with circadian, anti-inflammatory, metabolic modulating, and antiapoptotic effects (170–172). It can influence lipid metabolism in the body through various pathways, such as adipocyte lipolysis, fat deposition, brown adipose tissue growth, beige adipogenesis, and white adipose tissue browning, which in turn affect energy expenditure (Fig. 4) (173). Several animal studies have shown exogenous MT supplementation reduces fat deposition in mice (174–176). Surprisingly, the particular effects of MT on the kidney have recently been established in animal models of obesity. Studies have shown that MT has antioxidant effects on the kidneys of obese T2DM and Zucker diabetic rats by mechanisms including increased glutathione

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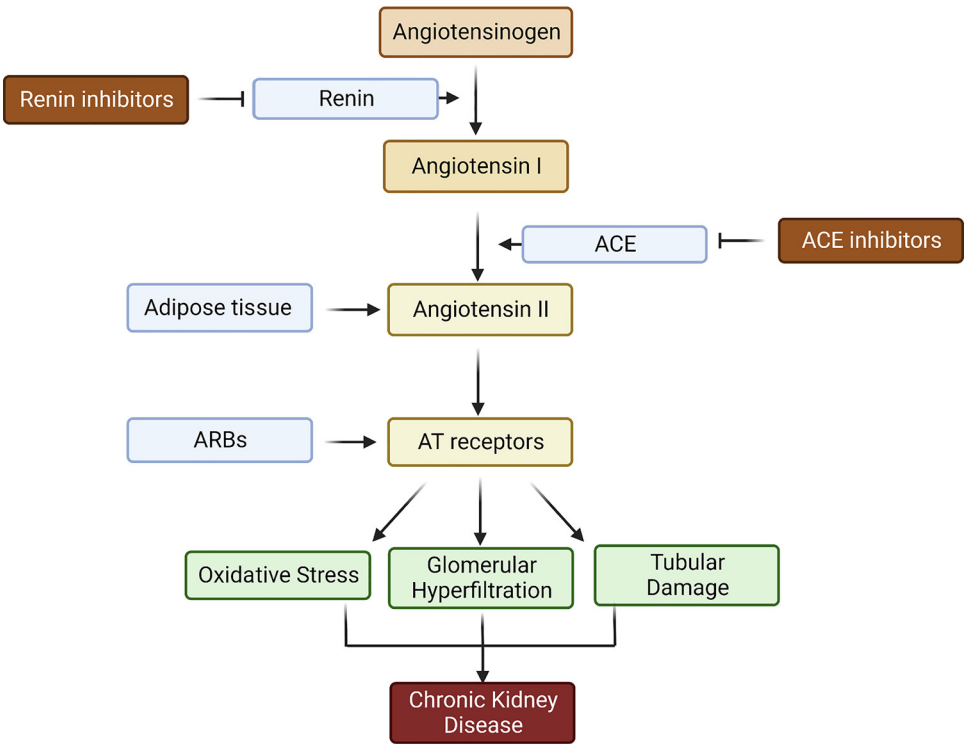

Figure 3. Role of the RAAS system and RAAS inhibitors in kidney injury. The RAAS is an antihypertensive regulatory system produced by the kidneys, causing vascular smooth muscle contraction and water and sodium retention, resulting in elevated blood pressure. Angiotensin I is converted into angiotensin II by an angiotensin-converting enzyme (ACE). Adipose tissue can induce angiotensin II secretion, which stimulates aldosterone secretion and promotes water and sodium reabsorption in renal tubules, improving the renal filtration rate and causing proteinuria. RAAS inhibitors reduce blood pressure and proteinuria and slow the decline of kidney function. RAAS, renin–angiotensin–aldosterone system. Created with BioRender.com. RAAS, renin-angiotensin-aldosterone system.

peroxidase activity, renal superoxide dismutase levels, and catalase activity (177, 178). Moreover, MT inhibits the formation of inflammatory vesicles associated with obesity-related nephropathy; thus, theoretically, MT is also protective against obesity-related nephropathy (179). However, there is still a long way to go before MT is actually used in clinical treatment, and this will require more effort on our part.

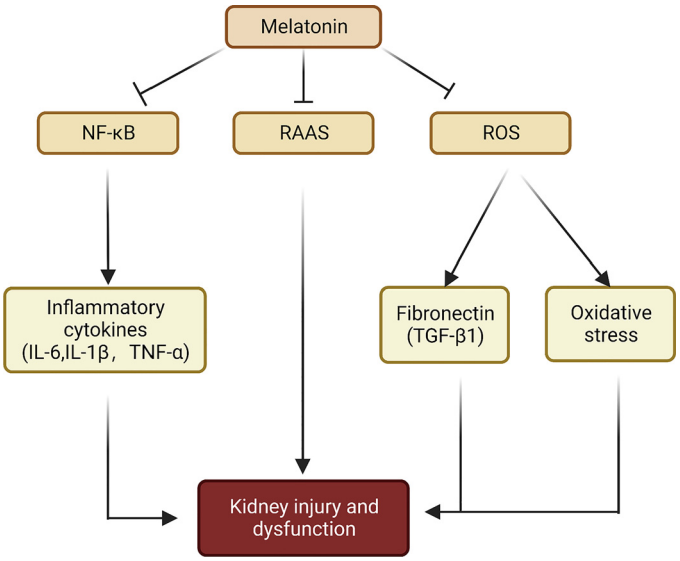

Figure 4. Melatonin in the treatment of kidney disease. Melatonin works on the kidney cells via melatonin receptors, and it has various pleiotropic properties, including antioxidant, anti-inflammation, and inhibition effects on the RAAS system. Melatonin, through antioxidant enzymes, can remove excess free radicals produced by mitochondria, reduce mitochondrial dysfunction, and act as an antioxidant. In addition, melatonin inhibits the RAAS system, reducing the glomerular filtration rate and proteinuria. Melatonin can also suppress transcription factors such as NF-κB, reduce the production of proinflammatory mediators, hence lowering inflammation, and effectively repair the kidney damage caused by obesity. Created with BioRender.com. RAAS, renin-angiotensin-aldosterone system.

CONCLUSIONS

The prevalence of CKD has increased with the rapid growth of the overweight and obese population in recent years, and obesity has become an independent risk factor for the development and progression of CKD and ESRD. Obesity has long been linked to kidney diseases such as glomerulopathies, nephrolithiasis, IgA nephropathy, and poor renal transplant survival. Although the exact mechanisms by which obesity causes or exacerbates nephropathy remain unclear, known possible mechanisms include multiple factors such as hemodynamic disorders, renal tissue hypoxia, IR, and RAAS activation. Further research is needed to characterize the pathophysiology and clinical spectrum of obesity-related nephropathy more fully. There is still a gap in the field for identifying early diagnostic markers of obesity-related CKD; although some experimental models have identified new markers that may contribute to the diagnosis of this disease, further research is needed before they can be used in clinical settings. In clinical practice, specific and effective treatments are lacking for obesity-related CKD. The main therapeutic interventions targeting patients' risk factors include weight loss, improvement of IR, and inhibition of RAAS activation to delay the deterioration of renal function. In addition, prebiotics, synbiotics, and MT may have protective effects against CKD; however, future specific clinical trials are needed to prove whether these drugs are effective in these patients. In conclusion, we still have many gaps to fill in the research on obesity-related CKD.

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DISCLOSURES

No conflicts of interest, financial or otherwise, are declared by the authors.

AUTHOR CONTRIBUTIONS

Z.J., X.Z., H.C., X.G., and G.W. conceived and designed research; Z.J., Y.W., M.H., X.R., and X.G. drafted manuscript; Z.J., Y.W., X.Z., H.C., M.H., X.R., X.G., and G.W. edited and revised manuscript; Z.J., Y.W., X.Z., H.C., M.H., X.R., X.G., and G.W. approved final version of manuscript.

REFERENCES

1. **González-Muniesa P**, **Mártinez-González MA**, **Hu FB**, **Després JP**, **Matsuzawa Y**, **Loos RJF**, **Moreno LA**, **Bray GA**, **Martinez JA.** Obesity. *Nat Rev Dis Primers* 3: 17034, 2017. doi:10.1038/nrdp.2017.34.
2. **Flegal KM**, **Kruszon-Moran D**, **Carroll MD**, **Fryar CD**, **Ogden CL.** Trends in obesity among adults in the United States, 2005 to 2014. *JAMA* 315: 2284–2291, 2016. doi:10.1001/jama.2016.6458.
3. **García-Carro C**, **Vergara A**, **Bermejo S**, **Azancot MA**, **Sellarés J**, **Soler MJ.** A nephrologist perspective on obesity: from kidney injury to clinical management. *Front Med (Lausanne)* 8: 655871, 2021. doi:10.3389/fmed.2021.655871.
4. **Wang Y**, **Xue H**, **Sun M**, **Zhu X**, **Zhao L**, **Yang Y.** Prevention and control of obesity in China. *Lancet Glob Health* 7: e1166–e1167, 2019. doi:10.1016/S2214-109X(19)30276-1.
5. **Haslam DW**, **James WP.** Obesity. *Lancet* 366: 1197–1209, 2005. doi:10.1016/S0140-6736(05)67483-1.
6. **Preble WE.** Obesity: observations on one thousand cases. *Boston Med Surg J* 188: 617–621, 1923. doi:10.1056/NEJM192304261881701.
7. **Hsu CY**, **McCulloch CE**, **Iribarren C**, **Darbinian J**, **Go AS.** Body mass index and risk for end-stage renal disease. *Ann Intern Med* 144: 21–28, 2006. doi:10.7326/0003-4819-144-1-200601030-00006.
8. **Iseki K**, **Ikemiya Y**, **Kinjo K**, **Inoue T**, **Iseki C**, **Takishita S.** Body mass index and the risk of development of end-stage renal disease in a screened cohort. *Kidney Int* 65: 1870–1876, 2004. doi:10.1111/j.1523-1755.2004.00582.x.
9. **Rutkowski P**, **Klassen A**, **Sebekova K**, **Bahner U**, **Heidland A.** Renal disease in obesity: the need for greater attention. *J Ren Nutr* 16: 216–223, 2006. doi:10.1053/j.jrn.2006.04.017.
10. **de Vries AP**, **Ruggenenti P**, **Ruan XZ**, **Praga M**, **Cruzado JM**, **Bajema IM**, **D'Agati VD**, **Lamb HJ**, **Pongrac Barlovic D**, **Hojs R**, **Abbate M**, **Rodriquez R**, **Mogensen CE**, **Porrini E**; **ERA-EDTA Working Group Diabesity.** Fatty kidney: emerging role of ectopic lipid in obesity-related renal disease. *Lancet Diabetes Endocrinol* 2: 417–426, 2014. doi:10.1016/S2213-8587(14)70065-8.
11. **D'Agati VD**, **Chagnac A**, **de Vries AP**, **Levi M**, **Porrini E**, **Herman-Edelstein M**, **Praga M.** Obesity-related glomerulopathy: clinical and pathologic characteristics and pathogenesis. *Nat Rev Nephrol* 12: 453–471, 2016. doi:10.1038/nrneph.2016.75.
12. **Wang M**, **Wang Z**, **Chen Y**, **Dong Y.** Kidney damage caused by obesity and its feasible treatment drugs. *Int J Mol Sci* 23: 747, 2022. doi:10.3390/ijms23020747.
13. **Wang Y**, **Chen X**, **Song Y**, **Caballero B**, **Cheskin LJ.** Association between obesity and kidney disease: a systematic review and meta-analysis. *Kidney Int* 73: 19–33, 2008. doi:10.1038/sj.ki.5002586.
14. **Kambham N**, **Markowitz GS**, **Valeri AM**, **Lin J**, **D'Agati VD.** Obesity-related glomerulopathy: an emerging epidemic. *Kidney Int* 59: 1498–1509, 2001. doi:10.1046/j.1523-1755.2001.0590041498.x.
15. **Hu R**, **Quan S**, **Wang Y**, **Zhou Y**, **Zhang Y**, **Liu L**, **Zhou XJ**, **Xing G.** Spectrum of biopsy proven renal diseases in Central China: a 10-year retrospective study based on 34,630 cases. *Sci Rep* 10: 10994, 2020. doi:10.1038/s41598-020-67910-w.
16. **Praga M**, **Hernández E**, **Morales E**, **Campos AP**, **Valero MA**, **Martínez MA**, **León M.** Clinical features and long-term outcome of obesity-associated focal segmental glomerulosclerosis. *Nephrol Dial Transplant* 16: 1790–1798, 2001. doi:10.1093/ndt/16.9.1790.
17. **Hall JE**, **Mouton AJ**, **da Silva AA**, **Omoto ACM**, **Wang Z**, **Li X**, **do Carmo JM.** Obesity, kidney dysfunction, and inflammation: interactions in hypertension. *Cardiovasc Res* 117: 1859–1876, 2021. doi:10.1093/cvr/cvaa336.
18. **Gai Z**, **Wang T**, **Visentin M**, **Kullak-Ublick GA**, **Fu X**, **Wang Z.** Lipid accumulation and chronic kidney disease. *Nutrients* 11: 722, 2019. doi:10.3390/nu11040722.
19. **Nicholl DDM**, **Ahmed SB**, **Loewen AHS**, **Hemmelgarn BR**, **Sola DY**, **Beecroft JM**, **Turin TC**, **Hanly PJ.** Declining kidney function increases the prevalence of sleep apnea and nocturnal hypoxia. *Chest* 141: 1422–1430, 2012. doi:10.1378/chest.11-1809.
20. **Maleki A**, **Montazeri M**, **Rashidi N**, **Montazeri M**, **Yousefi-Abdolmaleki E.** Metabolic syndrome and its components associated with chronic kidney disease. *J Res Med Sci* 20: 465–469, 2015. doi:10.4103/1735-1995.163969.
21. **Chen J**, **Gu D**, **Chen CS**, **Wu X**, **Hamm LL**, **Muntner P**, **Batuman V**, **Lee CH**, **Whelton PK**, **He J.** Association between the metabolic syndrome and chronic kidney disease in Chinese adults. *Nephrol Dial Transplant* 22: 1100–1106, 2007. doi:10.1093/ndt/gfl759.
22. **Zomorrodian D**, **Khajavi-Rad A**, **Avan A**, **Ebrahimi M**, **Nematy M**, **Azarpazhooh MR**, **Emamian M**, **Sadeghzade M**, **Mirhafez SR**, **Mohammadi M**, **Mousavi M**, **Esmaeili H**, **Moohebati M**, **Parizadeh MR**, **Ferns GA**, **Ghayour-Mobarhan M.** Metabolic syndrome components as markers to prognosticate the risk of developing chronic kidney disease: evidence-based study with 6492 individuals. *J Epidemiol Community Health* 69: 594–598, 2015. doi:10.1136/jech-2014-205160.
23. **Tsuboi N**, **Koike K**, **Hirano K**, **Utsunomiya Y**, **Kawamura T**, **Hosoya T.** Clinical features and long-term renal outcomes of Japanese patients with obesity-related glomerulopathy. *Clin Exp Nephrol* 17: 379–385, 2013. doi:10.1007/s10157-012-0719-y.
24. **Grillo MA**, **Mariani G**, **Ferraris JR.** Prematurity and low birth weight in neonates as a risk factor for obesity, hypertension, and chronic kidney disease in pediatric and adult age. *Front Med (Lausanne)* 8: 769734, 2021. doi:10.3389/fmed.2021.769734.
25. **Esmeijer K**, **de Vries AP**, **Mook-Kanamori DO**, **de Fijter JW**, **Rosendaal FR**, **Rabelink TJ**, **Smit RAJ**, **de Mutsert R**, **Hoogeveen EK.** Low birth weight and kidney function in middle-aged men and women: the Netherlands Epidemiology of Obesity Study. *Am J Kidney Dis* 74: 751–760, 2019. doi:10.1053/j.ajkd.2019.05.007.
26. **Silverwood RJ**, **Pierce M**, **Hardy R**, **Sattar N**, **Whincup P**, **Ferro C**, **Savage C**, **Kuh D**, **Nitsch D.** Low birth weight, later renal function, and the roles of adulthood blood pressure, diabetes, and obesity in a British birth cohort. *Kidney Int* 84: 1262–1270, 2013. doi:10.1038/ki.2013.223.
27. **González E**, **Gutiérrez E**, **Morales E**, **Hernández E**, **Andres A**, **Bello I**, **Díaz-González R**, **Leiva O**, **Praga M.** Factors influencing the progression of renal damage in patients with unilateral renal agenesis and remnant kidney. *Kidney Int* 68: 263–270, 2005. doi:10.1111/j.1523-1755.2005.00401.x.
28. **Abitbol CL**, **Chandar J**, **Rodríguez MM**, **Berho M**, **Seeherunvong W**, **Freundlich M**, **Zilleruelo G.** Obesity and preterm birth: additive risks in the progression of kidney disease in children. *Pediatr Nephrol* 24: 1363–1370, 2009. doi:10.1007/s00467-009-1120-2.
29. **Zhu P**, **Herrington WG**, **Haynes R**, **Emberson J**, **Landray MJ**, **Sudlow CLM**, **Woodward M**, **Baigent C**, **Lewington S**, **Staplin N.** Conventional and genetic evidence on the association between adiposity and CKD. *J Am Soc Nephrol* 32: 127–137, 2021. doi:10.1681/ASN.2020050679.

Downloaded from journals.physiology.org/journal/ajpendo at Univ of Chicago (128.135.204.136) on March 8, 2026.

30. **Wei L**, **Li Y**, **Yu Y**, **Xu M**, **Chen H**, **Li L**, **Peng T**, **Zhao K**, **Zhuang Y.** Obesity-related glomerulopathy: from mechanism to therapeutic target. *Diabetes Metab Syndr Obes* 14: 4371–4380, 2021. doi:10.2147/DMSO.S334199.
31. **Porrini E**, **Ruggenenti P**, **Luis-Lima S**, **Carrara F**, **Jiménez A**, **de Vries APJ**, **Torres A**, **Gaspari F**, **Remuzzi G.** Estimated GFR: time for a critical appraisal. *Nat Rev Nephrol* 15: 177–190, 2019 [Erratum in *Nat Rev Nephrol* 15: 121, 2018]. doi:10.1038/s41581-018-0080-9.
32. **López-Martínez M**, **Luis-Lima S**, **Morales E**, **Navarro-Díaz M**, **Negrín-Mena N**, **Folgueras T**, **Escamilla B**, **Estupiñán S**, **Delgado-Mallén P**, **Marrero-Miranda D**, **González-Rinne A**, **Miquel-Rodríguez RM**, **Cobo-Caso MA**, **Díaz-Martín L**, **Jiménez-Sosa A**, **González-Rinne F**, **Torres A**, **Porrini E.** The estimation of GFR and the adjustment for BSA in overweight and obesity: a dreadful combination of two errors. *Int J Obes (Lond)* 44: 1129–1140, 2020. doi:10.1038/s41366-019-0476-z.
33. **Zhang WR**, **Parikh CR.** Biomarkers of acute and chronic kidney disease. *Annu Rev Physiol* 81: 309–333, 2019. doi:10.1146/annurev-physiol-020518-114605.
34. **Mori Y**, **Ajay AK**, **Chang JH**, **Mou S**, **Zhao H**, **Kishi S**, **Li J**, **Brooks CR**, **Xiao S**, **Woo HM**, **Sabbisetti VS**, **Palmer SC**, **Galichon P**, **Li L**, **Henderson JM**, **Kuchroo VK**, **Hawkins J**, **Ichimura T**, **Bonventre JV.** KIM-1 mediates fatty acid uptake by renal tubular cells to promote progressive diabetic kidney disease. *Cell Metab* 33: 1042–1061.e7, 2021. doi:10.1016/j.cmet.2021.04.004.
35. **Safaeian B**, **Nickavar A**, **Zaeri H**, **Lahootian L**, **Behnampour N.** Utility of urine *N*-acetyl-β-D-glucosaminidase for prediction of renal damage in obese children. *Saudi J Kidney Dis Transpl* 32: 699–702, 2021. doi:10.4103/1319-2442.336764.
36. **Ding W**, **Mak RH.** Early markers of obesity-related renal injury in childhood. *Pediatr Nephrol* 30: 1–4, 2015. doi:10.1007/s00467-014-2976-3.
37. **Tofte N**, **Lindhardt M**, **Adamova K**, **Bakker SJL**, **Beige J**, **Beulens JWJ et al.** Early detection of diabetic kidney disease by urinary proteomics and subsequent intervention with spironolactone to delay progression (PRIORITY): a prospective observational study and embedded randomised placebo-controlled trial. *Lancet Diabetes Endocrinol* 8: 301–312, 2020. doi:10.1016/S2213-8587(20)30026-7.
38. **Pontillo C**, **Zhang ZY**, **Schanstra JP**, **Jacobs L**, **Zürbig P**, **Thijs L**, **Ramírez-Torres A**, **Heerspink HJL**, **Lindhardt M**, **Klein R**, **Orchard T**, **Porta M**, **Bilous RW**, **Charturvedi N**, **Rossing P**, **Vlahou A**, **Schepers E**, **Glorieux G**, **Mullen W**, **Delles C**, **Verhamme P**, **Vanholder R**, **Staessen JA**, **Mischak H**, **Jankowski J.** Prediction of chronic kidney disease stage 3 by CKD273, a urinary proteomic biomarker. *Kidney Int Rep* 2: 1066–1075, 2017. doi:10.1016/j.ekir.2017.06.004.
39. **Martinez-Montoro JI**, **Morales E**, **Cornejo-Pareja I**, **Tinahones FJ**, **Fernandez-Garcia JC.** Obesity-related glomerulopathy: current approaches and future perspectives. *Obes Rev* 23: e13450, 2022. doi:10.1111/obr.13450.
40. **Al Salmi I**, **Al Hajriy M**, **Hannawi S.** Ultrasound measurement and kidney development: a mini-review for nephrologists. *Saudi J Kidney Dis Transpl* 32: 174–182, 2021. doi:10.4103/1319-2442.318520.
41. **Gosmanova EO**, **Wu S**, **O'Neill WC.** Application of ultrasound in nephrology practice. *Adv Chronic Kidney Dis* 16: 396–404, 2009. doi:10.1053/j.ackd.2009.07.001.
42. **Foster MC**, **Hwang SJ**, **Porter SA**, **Massaro JM**, **Hoffmann U**, **Fox CS.** Development and reproducibility of a computed tomography-based measurement of renal sinus fat. *BMC Nephrol* 12: 52, 2011. doi:10.1186/1471-2369-12-52.
43. **Renzulli M**, **Brocchi S**, **Pettinari I**, **Biselli M**, **Clemente A**, **Corcioni B**, **Cappabianca S**, **Gaudiano C**, **Golfieri R.** New MRI series for kidney evaluation: saving time and money. *Br J Radiol* 92: 20190260, 2019. doi:10.1259/bjr.20190260.
44. **Nikken JJ**, **Krestin GP.** MRI of the kidney-state of the art. *Eur Radiol* 17: 2780–2793, 2007. doi:10.1007/s00330-007-0701-3.
45. **Weisinger JR**, **Kempson RL**, **Eldridge FL**, **Swenson RS.** The nephrotic syndrome: a complication of massive obesity. *Ann Intern Med* 81: 440–447, 1974. doi:10.7326/0003-4819-81-4-440.
46. **Serra A**, **Romero R**, **Lopez D**, **Navarro M**, **Esteve A**, **Perez N**, **Alastrue A**, **Ariza A.** Renal injury in the extremely obese patients with normal renal function. *Kidney Int* 73: 947–955, 2008. doi:10.1038/sj.ki.5002796.
47. **Choung HG**, **Bomback AS**, **Stokes MB**, **Santoriello D**, **Campenot ES**, **Batal I**, **Markowitz GS**, **D'Agati VD.** The spectrum of kidney biopsy findings in patients with morbid obesity. *Kidney Int* 95: 647–654, 2019. doi:10.1016/j.kint.2018.11.026.
48. **Bose B**, **Cattran D**; **Toronto Glomerulonephritis Registry.** Glomerular diseases: FSGS. *Clin J Am Soc Nephrol* 9: 626–632, 2014. doi:10.2215/CJN.05810513.
49. **Tsuboi N**, **Utsunomiya Y**, **Kanzaki G**, **Koike K**, **Ikegami M**, **Kawamura T**, **Hosoya T.** Low glomerular density with glomerulomegaly in obesity-related glomerulopathy. *Clin J Am Soc Nephrol* 7: 735–741, 2012. doi:10.2215/CJN.07270711.
50. **Tsuboi N**, **Utsunomiya Y**, **Hosoya T.** Obesity-related glomerulopathy and the nephron complement. *Nephrol Dial Transplant* 28, *Suppl* 4: iv108–iv113, 2013. doi:10.1093/ndt/gft258.
51. **Chen HM**, **Liu ZH**, **Zeng CH**, **Li SJ**, **Wang QW**, **Li LS.** Podocyte lesions in patients with obesity-related glomerulopathy. *Am J Kidney Dis* 48: 772–779, 2006. doi:10.1053/j.ajkd.2006.07.025.
52. **Kopp JB**, **Anders HJ**, **Susztak K**, **Podestà MA**, **Remuzzi G**, **Hildebrandt F**, **Romagnani P.** Podocytopathies. *Nat Rev Dis Primers* 6: 68, 2020. doi:10.1038/s41572-020-0196-7.
53. **Amann K**, **Benz K.** Structural renal changes in obesity and diabetes. *Semin Nephrol* 33: 23–33, 2013. doi:10.1016/j.semnephrol.2012.12.003.
54. **Kuwahara S**, **Hosojima M**, **Kaneko R**, **Aoki H**, **Nakano D**, **Sasagawa T**, **Kabasawa H**, **Kaseda R**, **Yasukawa R**, **Ishikawa T**, **Suzuki A**, **Sato H**, **Kageyama S**, **Tanaka T**, **Kitamura N**, **Narita I**, **Komatsu M**, **Nishiyama A**, **Saito A.** Megalin-mediated tubuloglomerular alterations in high-fat diet-induced kidney disease. *J Am Soc Nephrol* 27: 1996–2008, 2016. doi:10.1681/ASN.2015020190.
55. **Kong Y**, **Zhao X**, **Qiu M**, **Lin Y**, **Feng P**, **Li S**, **Liang B**, **Zhu Q**, **Huang H**, **Li C**, **Wang W.** Tubular Mas receptor mediates lipid-induced kidney injury. *Cell Death Dis* 12: 110, 2021. doi:10.1038/s41419-020-03375-z.
56. **Rodríguez RR**, **González-Bulnes A**, **Garcia-Contreras C**, **Elena Rodriguez-Rodriguez A**, **Astiz S**, **Vazquez-Gomez M**, **Luis Pesantez J**, **Isabel B**, **Salido-Ruiz E**, **González J**, **Donate Correa J**, **Luis-Lima S**, **Porrini E.** The Iberian pig fed with high-fat diet: a model of renal disease in obesity and metabolic syndrome. *Int J Obes (Lond)* 44: 457–465, 2020. doi:10.1038/s41366-019-0434-9.
57. **Gluba A**, **Mikhailidis DP**, **Lip GY**, **Hannam S**, **Rysz J**, **Banach M.** Metabolic syndrome and renal disease. *Int J Cardiol* 164: 141–150, 2013. doi:10.1016/j.ijcard.2012.01.013.
58. **Tobar A**, **Ori Y**, **Benchetrit S**, **Milo G**, **Herman-Edelstein M**, **Zingerman B**, **Lev N**, **Gafter U**, **Chagnac A.** Proximal tubular hypertrophy and enlarged glomerular and proximal tubular urinary space in obese subjects with proteinuria. *PLoS One* 8: e75547, 2013. doi:10.1371/journal.pone.0075547.
59. **Wuerzner G**, **Pruijm M**, **Maillard M**, **Bovet P**, **Renaud C**, **Burnier M**, **Bochud M.** Marked association between obesity and glomerular hyperfiltration: a cross-sectional study in an African population. *Am J Kidney Dis* 56: 303–312, 2010. doi:10.1053/j.ajkd.2010.03.017.
60. **Romagnani P**, **Remuzzi G**, **Glassock R**, **Levin A**, **Jager KJ**, **Tonelli M**, **Massy Z**, **Wanner C**, **Anders HJ.** Chronic kidney disease. *Nat Rev Dis Primers* 3: 17088, 2017. doi:10.1038/nrdp.2017.88.
61. **Chagnac A**, **Zingerman B**, **Rozen-Zvi B**, **Herman-Edelstein M.** Consequences of glomerular hyperfiltration: the role of physical forces in the pathogenesis of chronic kidney disease in diabetes and obesity. *Nephron* 143: 38–42, 2019. doi:10.1159/000499486.
62. **Fukuda A**, **Chowdhury MA**, **Venkatareddy MP**, **Wang SQ**, **Nishizono R**, **Suzuki T**, **Wickman LT**, **Wiggins JE**, **Muchayi T**, **Fingar D**, **Shedden KA**, **Inoki K**, **Wiggins RC.** Growth-dependent podocyte failure causes glomerulosclerosis. *J Am Soc Nephrol* 23: 1351–1363, 2012. doi:10.1681/ASN.2012030271.
63. **Mundel P**, **Shankland SJ.** Podocyte biology and response to injury. *J Am Soc Nephrol* 13: 3005–3015, 2002. doi:10.1097/01.asn.0000039661.06947.fd.
64. **Meyer-Schwesinger C.** The ins-and-outs of podocyte lipid metabolism. *Kidney Int* 98: 1087–1090, 2020. doi:10.1016/j.kint.2020.07.008.
65. **Wiggins JE**, **Goyal M**, **Sanden SK**, **Wharram BL**, **Shedden KA**, **Misek DE**, **Kuick RD**, **Wiggins RC.** Podocyte hypertrophy, "adaptation," and "decompensation" associated with glomerular enlargement and glomerulosclerosis in the aging rat: prevention by calorie restriction. *J Am Soc Nephrol* 16: 2953–2966, 2005. doi:10.1681/ASN.2005050488.

Downloaded from journals.physiology.org/journal/ajpendo at Univ of Chicago (128.135.204.136) on March 8, 2026.

66. **Deegens JK**, **Dijkman HB**, **Borm GF**, **Steenbergen EJ**, **van den Berg JG**, **Weening JJ**, **Wetzels JF.** Podocyte foot process effacement as a diagnostic tool in focal segmental glomerulosclerosis. *Kidney Int* 74: 1568–1576, 2008. doi:10.1038/ki.2008.413.
67. **Jiang T**, **Wang Z**, **Proctor G**, **Moskowitz S**, **Liebman SE**, **Rogers T**, **Lucia MS**, **Li J**, **Levi M.** Diet-induced obesity in C57BL/6J mice causes increased renal lipid accumulation and glomerulosclerosis via a sterol regulatory element-binding protein-1c-dependent pathway. *J Biol Chem* 280: 32317–32325, 2005. doi:10.1074/jbc.M500801200.
68. **D'Agati VD**, **Kaskel FJ**, **Falk RJ.** Focal segmental glomerulosclerosis. *N Engl J Med* 365: 2398–2411, 2011. doi:10.1056/NEJMra1106556.
69. **D'Agati V.** Pathologic classification of focal segmental glomerulosclerosis. *Semin Nephrol* 23: 117–134, 2003. doi:10.1053/snep.2003.50012.
70. Hyperfiltration in remnant nephrons: a potentially adverse response to renal ablation. *J Am Soc Nephrol* 12: 1315–1325, 2001. doi:10.1681/ASN.V1261315.
71. **Praga M**, **Hernández E**, **Herrero JC**, **Morales E**, **Revilla Y**, **Díaz-González R**, **Rodicio JL.** Influence of obesity on the appearance of proteinuria and renal insufficiency after unilateral nephrectomy. *Kidney Int* 58: 2111–2118, 2000. doi:10.1111/j.1523-1755.2000.00384.x.
72. **Praga M**, **Morales E.** Obesity, proteinuria and progression of renal failure. *Curr Opin Nephrol Hypertens* 15: 481–486, 2006. doi:10.1097/01.mnh.0000242172.06459.7c.
73. **Li LC**, **Yang JL**, **Lee WC**, **Chen JB**, **Lee CT**, **Wang PW**, **Vaghese Z**, **Chen WY.** Palmitate aggravates proteinuria-induced cell death and inflammation via CD36-inflammasome axis in the proximal tubular cells of obese mice. *Am J Physiol Renal Physiol* 315: F1720–F1731, 2018. doi:10.1152/ajprenal.00536.2017.
74. **Ding XQ**, **Jian TY**, **Gai YN**, **Niu GT**, **Liu Y**, **Meng XH**, **Li J**, **Lyu H**, **Ren BR**, **Chen J.** Chicoric acid attenuated renal tubular injury in HFD-induced chronic kidney disease mice through the promotion of mitophagy via the Nrf2/PINK/parkin pathway. *J Agric Food Chem* 70: 2923–2935, 2022. doi:10.1021/acs.jafc.1c07795.
75. **Hallow KM**, **Gebremichael Y**, **Helmlinger G**, **Vallon V.** Primary proximal tubule hyperreabsorption and impaired tubular transport counterregulation determine glomerular hyperfiltration in diabetes: a modeling analysis. *Am J Physiol Renal Physiol* 312: F819–F835, 2017. doi:10.1152/ajprenal.00497.2016.
76. **Hall JE.** Mechanisms of abnormal renal sodium handling in obesity hypertension. *Am J Hypertens* 10: 49S–55S, 1997.
77. **Hall JE**, **Kuo JJ**, **da Silva AA**, **de Paula RB**, **Liu J**, **Tallam L.** Obesity-associated hypertension and kidney disease. *Curr Opin Nephrol Hypertens* 12: 195–200, 2003. doi:10.1097/00041552-200303000-00011.
78. **Nordquist L**, **Brown R**, **Fasching A**, **Persson P**, **Palm F.** Proinsulin C-peptide reduces diabetes-induced glomerular hyperfiltration via efferent arteriole dilation and inhibition of tubular sodium reabsorption. *Am J Physiol Renal Physiol* 297: F1265–F1272, 2009. doi:10.1152/ajprenal.00228.2009.
79. **Fukushima K**, **Kitamura S**, **Tsuji K**, **Sang Y**, **Wada J.** Sodium glucose co-transporter 2 inhibitor ameliorates autophagic flux impairment on renal proximal tubular cells in obesity mice. *Int J Mol Sci* 21: 4054, 2020. doi:10.3390/ijms21114054.
80. **Hall JE.** The kidney, hypertension, and obesity. *Hypertension* 41: 625–633, 2003. doi:10.1161/01.HYP.0000052314.95497.78.
81. **Tsuboi N**, **Okabayashi Y.** The renal pathology of obesity: structure-function correlations. *Semin Nephrol* 41: 296–306, 2021. doi:10.1016/j.semnephrol.2021.06.002.
82. **Chang Y**, **Ryu S**, **Choi Y**, **Zhang Y**, **Cho J**, **Kwon MJ**, **Hyun YY**, **Lee KB**, **Kim H**, **Jung HS**, **Yun KE**, **Ahn J**, **Rampal S**, **Zhao D**, **Suh BS**, **Chung EC**, **Shin H**, **Pastor-Barriuso R**, **Guallar E.** Metabolically healthy obesity and development of chronic kidney disease: a cohort study. *Ann Intern Med* 164: 305–312, 2016. doi:10.7326/M15-1323.
83. **Cho YK**, **Lee J**, **Kim HS**, **Park JY**, **Lee WJ**, **Kim YJ**, **Jung CH.** Impact of transition in metabolic health and obesity on the incident chronic kidney disease: a nationwide cohort study. *J Clin Endocrinol Metab* 105: dgaa033, 2020. doi:10.1210/clinem/dgaa033.
84. **Cao X**, **Zhou J**, **Yuan H**, **Wu L**, **Chen Z.** Chronic kidney disease among overweight and obesity with and without metabolic syndrome in an urban Chinese cohort. *BMC Nephrol* 16: 85, 2015. doi:10.1186/s12882-015-0083-8.
85. **Wahba IM**, **Mak RH.** Obesity and obesity-initiated metabolic syndrome: mechanistic links to chronic kidney disease. *Clin J Am Soc Nephrol* 2: 550–562, 2007. doi:10.2215/CJN.04071206.
86. **Beddhu S.** The body mass index paradox and an obesity, inflammation, and atherosclerosis syndrome in chronic kidney disease. *Semin Dial* 17: 229–232, 2004. doi:10.1111/j.0894-0959.2004.17311.x.
87. **Wellen KE**, **Hotamisligil GS.** Obesity-induced inflammatory changes in adipose tissue. *J Clin Invest* 112: 1785–1788, 2003. doi:10.1172/JCI20514.
88. **Villarroya F**, **Cereijo R**, **Gavaldà-Navarro A**, **Villarroya J**, **Giralt M.** Inflammation of brown/beige adipose tissues in obesity and metabolic disease. *J Intern Med* 284: 492–504, 2018. doi:10.1111/joim.12803.
89. **Declèves AE**, **Sharma K.** Obesity and kidney disease: differential effects of obesity on adipose tissue and kidney inflammation and fibrosis. *Curr Opin Nephrol Hypertens* 24: 28–36, 2015. doi:10.1097/MNH.0000000000000087.
90. **Ruiz-Ortega M**, **Rayego-Mateos S**, **Lamas S**, **Ortiz A**, **Rodrigues-Diez RR.** Targeting the progression of chronic kidney disease. *Nat Rev Nephrol* 16: 269–288, 2020. doi:10.1038/s41581-019-0248-y.
91. **Sun Y**, **Ge X**, **Li X**, **He J**, **Wei X**, **Du J**, **Sun J**, **Li X**, **Xun Z**, **Liu W**, **Zhang H**, **Wang ZY**, **Li YC.** High-fat diet promotes renal injury by inducing oxidative stress and mitochondrial dysfunction. *Cell Death Dis* 11: 914, 2020. doi:10.1038/s41419-020-03122-4.
92. **Pierine DT**, **Navarro ME**, **Minatel IO**, **Luvizotto RA**, **Nascimento AF**, **Ferreira AL**, **Yeum KJ**, **Corrêa CR.** Lycopene supplementation reduces TNF-α via RAGE in the kidney of obese rats. *Nutr Diabetes* 4: e142, 2014. doi:10.1038/nutd.2014.39.
93. **Kanda H**, **Tateya S**, **Tamori Y**, **Kotani K**, **Hiasa K**, **Kitazawa R**, **Kitazawa S**, **Miyachi H**, **Maeda S**, **Egashira K**, **Kasuga M.** MCP-1 contributes to macrophage infiltration into adipose tissue, insulin resistance, and hepatic steatosis in obesity. *J Clin Invest* 116: 1494–1505, 2006. doi:10.1172/JCI26498.
94. **Stemmer K**, **Perez-Tilve D**, **Ananthakrishnan G**, **Bort A**, **Seeley RJ**, **Tschöp MH**, **Dietrich DR**, **Pfluger PT.** High-fat-diet-induced obesity causes an inflammatory and tumor-promoting microenvironment in the rat kidney. *Dis Model Mech* 5: 627–635, 2012. doi:10.1242/dmm.009407.
95. **Gotoh K**, **Inoue M**, **Masaki T**, **Chiba S**, **Shiraishi K**, **Shimasaki T**, **Matsuoka K**, **Ando H**, **Fujiwara K**, **Fukunaga N**, **Aoki K**, **Nawata T**, **Katsuragi I**, **Kakuma T**, **Seike M**, **Yoshimatsu H.** Obesity-related chronic kidney disease is associated with spleen-derived IL-10. *Nephrol Dial Transplant* 28: 1120–1130, 2013. doi:10.1093/ndt/gfs440.
96. **Wang H**, **Li J**, **Gai Z**, **Kullak-Ublick GA**, **Liu Z.** TNF-α deficiency prevents renal inflammation and oxidative stress in obese mice. *Kidney Blood Press Res* 42: 416–427, 2017. doi:10.1159/000478869.
97. **Browne T**, **Forfang D**, **Bethel S**, **Joseph J**, **Brereton L**, **Damron KC.** The National Kidney Foundation's efforts to improve and increase patient-centered CKD research. *Am J Kidney Dis* 77: 471–473, 2021. doi:10.1053/j.ajkd.2020.11.008.
98. **Griffin KA**, **Kramer H**, **Bidani AK.** Adverse renal consequences of obesity. *Am J Physiol Renal Physiol* 294: F685–F696, 2008. doi:10.1152/ajprenal.00324.2007.
99. **Reynolds K**, **Gu D**, **Muntner P**, **Kusek JW**, **Chen J**, **Wu X**, **Duan X**, **Chen CS**, **Klag MJ**, **Whelton PK**, **He J.** A population-based, prospective study of blood pressure and risk for end-stage renal disease in China. *J Am Soc Nephrol* 18: 1928–1935, 2007. doi:10.1681/ASN.2006111199.
100. **Bobulescu IA.** Renal lipid metabolism and lipotoxicity. *Curr Opin Nephrol Hypertens* 19: 393–402, 2010. doi:10.1097/MNH.0b013e32833aa4ac.
101. **Rickards E.** Remarks on the fatty transformation of the kidney. *Br Med J* 2: 2–3, 1883. doi:10.1136/bmj.2.1175.2.
102. **Moorhead JF**, **Chan MK**, **El-Nahas M**, **Varghese Z.** Lipid nephrotoxicity in chronic progressive glomerular and tubulo-interstitial disease. *Lancet* 2: 1309–1311, 1982. doi:10.1016/s0140-6736(82)91513-6.
103. **Zhang YB**, **Sheng LT**, **Wei W**, **Guo H**, **Yang H**, **Min X**, **Guo K**, **Yang K**, **Zhang X**, **He M**, **Wu T**, **Pan A.** Association of blood lipid profile with incident chronic kidney disease: a Mendelian randomization study. *Atherosclerosis* 300: 19–25, 2020. doi:10.1016/j.atherosclerosis.2020.03.020.
104. **Weldegiorgis M**, **Woodward M.** Elevated triglycerides and reduced high-density lipoprotein cholesterol are independently associated with the onset of advanced chronic kidney disease: a cohort study of 911,360 individuals from the United Kingdom. *BMC Nephrol* 23:

Downloaded from journals.physiology.org/journal/ajpendo at Univ of Chicago (128.135.204.136) on March 8, 2026.

312, 2022 [Erratum in *BMC Nephrol* 23: 318, 2022]. doi:10.1186/s12882-022-02932-2.

105. **Lanktree MB**, **Thériault S**, **Walsh M**, **Paré G.** HDL cholesterol, LDL cholesterol, and triglycerides as risk factors for CKD: a Mendelian Randomization Study. *Am J Kidney Dis* 71: 166–172, 2018. doi:10.1053/j.ajkd.2017.06.011.
106. **Hammoud SH**, **AlZaim I**, **Al-Dhaheri Y**, **Eid AH**, **El-Yazbi AF.** Perirenal adipose tissue inflammation: novel insights linking metabolic dysfunction to renal diseases. *Front Endocrinol (Lausanne)* 12: 707126, 2021. doi:10.3389/fendo.2021.707126.
107. **Huang N**, **Mao E-W**, **Hou N-N**, **Liu Y-P**, **Han F**, **Sun X-D.** Novel insight into perirenal adipose tissue: a neglected adipose depot linking cardiovascular and chronic kidney disease. *World J Diabetes* 11: 115–125, 2020. doi:10.4239/wjd.v11.i4.115.
108. **Foster MC**, **Hwang SJ**, **Porter SA**, **Massaro JM**, **Hoffmann U**, **Fox CS.** Fatty kidney, hypertension, and chronic kidney disease: the Framingham Heart Study. *Hypertension* 58: 784–790, 2011. doi:10.1161/HYPERTENSIONAHA.111.175315.
109. **Klausen KP**, **Parving HH**, **Scharling H**, **Jensen JS.** Microalbuminuria and obesity: impact on cardiovascular disease and mortality. *Clin Endocrinol (Oxf)* 71: 40–45, 2009. doi:10.1111/j.1365-2265.2008.03427.x.
110. **D'Marco L**, **Salazar J**, **Cortez M**, **Salazar M**, **Wettel M**, **Lima-Martínez M**, **Rojas E**, **Roque W**, **Bermúdez V.** Perirenal fat thickness is associated with metabolic risk factors in patients with chronic kidney disease. *Kidney Res Clin Pract* 38: 365–372, 2019. doi:10.23876/j.krcp.18.0155.
111. **Liu BX**, **Sun W**, **Kong XQ.** Perirenal fat: a unique fat pad and potential target for cardiovascular disease. *Angiology* 70: 584–593, 2019. doi:10.1177/0003319718799967.
112. **Ma S**, **Zhu XY**, **Eirin A**, **Woollard JR**, **Jordan KL**, **Tang H**, **Lerman A**, **Lerman LO.** Perirenal fat promotes renal arterial endothelial dysfunction in obese swine through tumor necrosis factor-α. *J Urol* 195: 1152–1159, 2016. doi:10.1016/j.juro.2015.08.105.
113. **Yoda-Murakami M**, **Taniguchi M**, **Takahashi K**, **Kawamata S**, **Saito K**, **Choi-Miura NH**, **Tomita M.** Change in expression of GBP28/adiponectin in carbon tetrachloride-administrated mouse liver. *Biochem Biophys Res Commun* 285: 372–377, 2001. doi:10.1006/bbrc.2001.5134.
114. **Patel JV**, **Abraheem A**, **Dotsenko O**, **Creamer J**, **Gunning M**, **Hughes EA**, **Lip GY.** Circulating serum adiponectin levels in patients with coronary artery disease: relationship to atherosclerotic burden and cardiac function. *J Intern Med* 264: 593–598, 2008. doi:10.1111/j.1365-2796.2008.02007.x.
115. **Martinez-Huenchullan SF**, **Tam CS**, **Ban LA**, **Ehrenfeld-Slater P**, **McLennan SV**, **Twigg SM.** Skeletal muscle adiponectin induction in obesity and exercise. *Metabolism* 102: 154008, 2020. doi:10.1016/j.metabol.2019.154008.
116. **Zhao S**, **Kusminski CM**, **Scherer PE.** Adiponectin, leptin and cardiovascular disorders. *Circ Res* 128: 136–149, 2021. doi:10.1161/CIRCRESAHA.120.314458.
117. **Kadowaki T**, **Yamauchi T.** Adiponectin and adiponectin receptors. *Endocr Rev* 26: 439–451, 2005. doi:10.1210/er.2005-0005.
118. **Kadowaki T**, **Yamauchi T**, **Kubota N**, **Hara K**, **Ueki K**, **Tobe K.** Adiponectin and adiponectin receptors in insulin resistance, diabetes, and the metabolic syndrome. *J Clin Invest* 116: 1784–1792, 2006. doi:10.1172/JCI29126.
119. **Yamauchi T**, **Kamon J**, **Waki H**, **Terauchi Y**, **Kubota N**, **Hara K**, **Mori Y**, **Ide T**, **Murakami K**, **Tsuboyama-Kasaoka N**, **Ezaki O**, **Akanuma Y**, **Gavrilova O**, **Vinson C**, **Reitman ML**, **Kagechika H**, **Shudo K**, **Yoda M**, **Nakano Y**, **Tobe K**, **Nagai R**, **Kimura S**, **Tomita M**, **Froguel P**, **Kadowaki T.** The fat-derived hormone adiponectin reverses insulin resistance associated with both lipoatrophy and obesity. *Nat Med* 7: 941–946, 2001. doi:10.1038/90984.
120. **Przybyciński J**, **Dziedziejko V**, **Puchałowicz K**, **Domański L**, **Pawlik A.** Adiponectin in chronic kidney disease. *Int J Mol Sci* 21: 9375, 2020. doi:10.3390/ijms21249375.
121. **Ohashi K**, **Iwatani H**, **Kihara S**, **Nakagawa Y**, **Komura N**, **Fujita K**, **Maeda N**, **Nishida M**, **Katsube F**, **Shimomura I**, **Ito T**, **Funahashi T.** Exacerbation of albuminuria and renal fibrosis in subtotal renal ablation model of adiponectin-knockout mice. *Arterioscler Thromb Vasc Biol* 27: 1910–1917, 2007. doi:10.1161/ATVBAHA.107.147645.
122. **Meyvis K**, **Verrijken A**, **Wouters K**, **Van Gaal L.** Plasma adiponectin level is inversely correlated with albuminuria in overweight and obese nondiabetic individuals. *Metabolism* 62: 1570–1576, 2013. doi:10.1016/j.metabol.2013.05.023.
123. **Yano Y**, **Hoshide S**, **Ishikawa J**, **Hashimoto T**, **Eguchi K**, **Shimada K**, **Kario K.** Differential impacts of adiponectin on low-grade albuminuria between obese and nonobese persons without diabetes. *J Clin Hypertens (Greenwich)* 9: 775–782, 2007. doi:10.1111/j.1524-6175.2007.07321.x.
124. **Pereira S**, **Cline DL**, **Glavas MM**, **Covey SD**, **Kieffer TJ.** Tissue-specific effects of leptin on glucose and lipid metabolism. *Endocr Rev* 42: 1–28, 2021. doi:10.1210/endrev/bnaa027.
125. **Zhang F**, **Chen Y**, **Heiman M**, **Dimarchi R.** Leptin: structure, function and biology. *Vitam Horm* 71: 345–372, 2005. doi:10.1016/S0083-6729(05)71012-8.
126. **Garibotto G**, **Russo R**, **Franceschini R**, **Robaudo C**, **Saffioti S**, **Sofia A**, **Rolandi E**, **Deferrari G**, **Barreca T.** Inter-organ leptin exchange in humans. *Biochem Biophys Res Commun* 247: 504–509, 1998. doi:10.1006/bbrc.1998.8819.
127. **Briley LP**, **Szczech LA.** Leptin and renal disease. *Semin Dial* 19: 54–59, 2006. doi:10.1111/j.1525-139X.2006.00119.x.
128. **Teixeira TM**, **da Costa DC**, **Resende AC**, **Soulage CO**, **Bezerra FF**, **Daleprane JB.** Activation of Nrf2-antioxidant signaling by 1,25-dihydroxycholecalciferol prevents leptin-induced oxidative stress and inflammation in human endothelial cells. *J Nutr* 147: 506–513, 2017. doi:10.3945/jn.116.239475.
129. **Considine RV.** Human leptin: an adipocyte hormone with weight-regulatory and endocrine functions. *Semin Vasc Med* 5: 15–24, 2005. doi:10.1055/s-2005-871738.
130. **Considine RV.** Increased serum leptin indicates leptin resistance in obesity. *Clin Chem* 57: 1461–1462, 2011. doi:10.1373/clinchem.2011.166421.
131. **Considine RV**, **Sinha MK**, **Heiman ML**, **Kriauciunas A**, **Stephens TW**, **Nyce MR**, **Ohannesian JP**, **Marco CC**, **McKee LJ**, **Bauer TL**, **Caro JF.** Serum immunoreactive-leptin concentrations in normal-weight and obese humans. *N Engl J Med* 334: 292–295, 1996. doi:10.1056/NEJM199602013340503.
132. **Han DC**, **Isono M**, **Chen S**, **Casaretto A**, **Hong SW**, **Wolf G**, **Ziyadeh FN.** Leptin stimulates type I collagen production in db/db mesangial cells: glucose uptake and TGF-β type II receptor expression. *Kidney Int* 59: 1315–1323, 2001. doi:10.1046/j.1523-1755.2001.0590041315.x.
133. **Kokot F**, **Wiecek A**, **Mesjasz J**, **Adamczak M**, **Spiechowicz U.** Influence of long-term recombinant human erythropoietin (rHuEpo) therapy on plasma leptin and neuropeptide Y concentration in haemodialysed uraemic patients. *Nephrol Dial Transplant* 13: 1200–1205, 1998. doi:10.1093/ndt/13.5.1200.
134. **Lu H**, **Cassis LA**, **Kooi CW**, **Daugherty A.** Structure and functions of angiotensinogen. *Hypertens Res* 39: 492–500, 2016 [Erratum in *Hypertens Res* 39: 827, 2016]. doi:10.1038/hr.2016.17.
135. **Sharma I**, **Liao Y**, **Zheng X**, **Kanwar YS.** New pandemic: obesity and associated nephropathy. *Front Med (Lausanne)* 8: 673556, 2021. doi:10.3389/fmed.2021.673556.
136. **Frigolet ME**, **Torres N**, **Tovar AR.** The renin-angiotensin system in adipose tissue and its metabolic consequences during obesity. *J Nutr Biochem* 24: 2003–2015, 2013. doi:10.1016/j.jnutbio.2013.07.002.
137. **Fischer H**, **Weiss RE**, **Friedman AN**, **Imam TH**, **Coleman KJ.** The relationship between kidney function and body mass index before and after bariatric surgery in patients with chronic kidney disease. *Surg Obes Relat Dis* 17: 508–515, 2021. doi:10.1016/j.soard.2020.11.010.
138. **Fourkiotis VG**, **Hanslik G**, **Hanusch F**, **Lepenies J**, **Quinkler M.** Aldosterone and the kidney. *Horm Metab Res* 44: 194–201, 2012. doi:10.1055/s-0031-1295461.
139. **Cao Z**, **Bonnet F**, **Davis B**, **Allen TJ**, **Cooper ME.** Additive hypotensive and anti-albuminuric effects of angiotensin-converting enzyme inhibition and angiotensin receptor antagonism in diabetic spontaneously hypertensive rats. *Clin Sci (Lond)* 100: 591–599, 2001.
140. **Bakris GL**, **Agarwal R**, **Anker SD**, **Pitt B**, **Ruilope LM**, **Rossing P**, **Kolkhof P**, **Nowack C**, **Schloemer P**, **Joseph A**, **Filippatos G**; **FIDELIO-DKD Investigators.** Effect of finerenone on chronic kidney disease outcomes in type 2 diabetes. *N Engl J Med* 383: 2219–2229, 2020. doi:10.1056/NEJMoa2025845.
141. **Kanasaki K**, **Kitada M**, **Kanasaki M**, **Koya D.** The biological consequence of obesity on the kidney. *Nephrol Dial Transplant* 28, *Suppl* 4: iv1–iv7, 2013. doi:10.1093/ndt/gft098.

Downloaded from journals.physiology.org/journal/ajpendo at Univ of Chicago (128.135.204.136) on March 8, 2026.

142. **Artunc F**, **Schleicher E**, **Weigert C**, **Fritsche A**, **Stefan N**, **Häring HU.** The impact of insulin resistance on the kidney and vasculature. *Nat Rev Nephrol* 12: 721–737, 2016. doi:10.1038/nrneph.2016.145.
143. **Sarafidis PA**, **Ruilope LM.** Insulin resistance, hyperinsulinemia, and renal injury: mechanisms and implications. *Am J Nephrol* 26: 232–244, 2006. doi:10.1159/000093632.
144. **Rao A**, **Pandya V**, **Whaley-Connell A.** Obesity and insulin resistance in resistant hypertension: implications for the kidney. *Adv Chronic Kidney Dis* 22: 211–217, 2015. doi:10.1053/j.ackd.2014.12.004.
145. **Lakkis JI**, **Weir MR.** Obesity and kidney disease. *Prog Cardiovasc Dis* 61: 157–167, 2018. doi:10.1016/j.pcad.2018.07.005.
146. **Chonchol M**, **Scragg R.** 25-Hydroxyvitamin D, insulin resistance, and kidney function in the Third National Health and Nutrition Examination Survey. *Kidney Int* 71: 134–139, 2007. doi:10.1038/sj.ki.5002002.
147. **Mykkänen L**, **Zaccaro DJ**, **Wagenknecht LE**, **Robbins DC**, **Gabriel M**, **Haffner SM.** Microalbuminuria is associated with insulin resistance in nondiabetic subjects: the insulin resistance atherosclerosis study. *Diabetes* 47: 793–800, 1998. doi:10.2337/diabetes.47.5.793.
148. **de Vinuesa SG**, **Goicoechea M**, **Kanter J**, **Puerta M**, **Cachofeiro V**, **Lahera V**, **Gómez-Campderá F**, **Luño J.** Insulin resistance, inflammatory biomarkers, and adipokines in patients with chronic kidney disease: effects of angiotensin II blockade. *J Am Soc Nephrol* 17: S206–S212, 2006. doi:10.1681/ASN.2006080916.
149. **Zhang Y**, **Lee ET**, **Howard BV**, **Best LG**, **Umans JG**, **Yeh J**, **Wang W**, **Yeh F**, **Ali T**, **Devereux RB**, **de Simone G.** Insulin resistance, incident cardiovascular diseases, and decreased kidney function among nondiabetic American Indians: the Strong Heart Study. *Diabetes care* 36: 3195–3200, 2013. doi:10.2337/dc12-2368.
150. **Gomes AC**, **Hoffmann C**, **Mota JF.** The human gut microbiota: metabolism and perspective in obesity. *Gut Microbes* 9: 308–325, 2018. doi:10.1080/19490976.2018.1465157.
151. **Matsumoto M**, **Kibe R**, **Ooga T**, **Aiba Y**, **Kurihara S**, **Sawaki E**, **Koga Y**, **Benno Y.** Impact of intestinal microbiota on intestinal luminal metabolome. *Sci Rep* 2: 233, 2012. doi:10.1038/srep00233.
152. **Nallu A**, **Sharma S**, **Ramezani A**, **Muralidharan J**, **Raj D.** Gut microbiome in chronic kidney disease: challenges and opportunities. *Transl Res* 179: 24–37, 2017. doi:10.1016/j.trsl.2016.04.007.
153. **Lovre D**, **Shah S**, **Sihota A**, **Fonseca VA.** Managing diabetes and cardiovascular risk in chronic kidney disease patients. *Endocrinol Metab Clin North Am* 47: 237–257, 2018. doi:10.1016/j.ecl.2017.10.006.
154. **López-Contreras BE**, **Morán-Ramos S**, **Villarruel-Vázquez R**, **Macías-Kauffer L**, **Villamil-Ramírez H**, **León-Mimila P**, **Vega-Badillo J**, **Sánchez-Muñoz F**, **Llanos-Moreno LE**, **Canizalez-Román A**, **Del Río-Navarro B**, **Ibarra-González I**, **Vela-Amieva M**, **Villarreal-Molina T**, **Ochoa-Leyva A**, **Aguilar-Salinas CA**, **Canizales-Quinteros S.** Composition of gut microbiota in obese and normal-weight Mexican school-age children and its association with metabolic traits. *Pediatr Obes* 13: 381–388, 2018. doi:10.1111/ijpo.12262.
155. **Popkov VA**, **Zharikova AA**, **Demchenko EA**, **Andrianova NV**, **Zorov DB**, **Plotnikov EY.** Gut microbiota as a source of uremic toxins. *Int J Mol Sci* 23: 483, 2022. doi:10.3390/ijms23010483.
156. **Wu IW**, **Hsu KH**, **Lee CC**, **Sun CY**, **Hsu HJ**, **Tsai CJ**, **Tzen CY**, **Wang YC**, **Lin CY**, **Wu MS.** p-Cresyl sulphate and indoxyl sulphate predict progression of chronic kidney disease. *Nephrol Dial Transplant* 26: 938–947, 2011. doi:10.1093/ndt/gfq580.
157. **Ramezani A**, **Raj DS.** The gut microbiome, kidney disease, and targeted interventions. *J Am Soc Nephrol* 25: 657–670, 2014. doi:10.1681/ASN.2013080905.
158. **Rysz J**, **Franczyk B**, **Ławiński J**, **Olszewski R**, **Ciałkowska-Rysz A**, **Gluba-Brzózka A.** The impact of CKD on uremic toxins and gut microbiota. *Toxins (Basel)* 13: 252, 2021. doi:10.3390/toxins13040252.
159. **Ramezani A**, **Massy ZA**, **Meijers B**, **Evenepoel P**, **Vanholder R**, **Raj DS.** Role of the Gut Microbiome in Uremia: A Potential Therapeutic Target. *Am J Kidney Dis* 67: 483–498, 2016. doi:10.1053/j.ajkd.2015.09.027.
160. **Antza C**, **Stabouli S**, **Kotsis V.** Gut microbiota in kidney disease and hypertension. *Pharmacol Res* 130: 198–203, 2018. doi:10.1016/j.phrs.2018.02.028.
161. **Rukavina Mikusic NL**, **Kouyoumdzian NM**, **Choi MR.** Gut microbiota and chronic kidney disease: evidences and mechanisms that mediate a new communication in the gastrointestinal-renal axis. *Pflugers Arch* 472: 303–320, 2020. doi:10.1007/s00424-020-02352-x.
162. **Borges NA**, **Barros AF**, **Nakao LS**, **Dolenga CJ**, **Fouque D**, **Mafra D.** Protein-bound uremic toxins from gut microbiota and inflammatory markers in chronic kidney disease. *J Ren Nutr* 26: 396–400, 2016. doi:10.1053/j.jrn.2016.07.005.
163. **Dalton TP**, **Puga A**, **Shertzer HG.** Induction of cellular oxidative stress by aryl hydrocarbon receptor activation. *Chem Biol Interact* 141: 77–95, 2002. doi:10.1016/s0009-2797(02)00067-4.
164. **Yang HL**, **Feng P**, **Xu Y**, **Hou YY**, **Ojo O**, **Wang XH.** The role of dietary fiber supplementation in regulating uremic toxins in patients with chronic kidney disease: a meta-analysis of randomized controlled trials. *J Ren Nutr* 31: 438–447, 2021. doi:10.1053/j.jrn.2020.11.008.
165. **Wu Y**, **Liu Z**, **Xiang Z**, **Zeng C**, **Chen Z**, **Ma X**, **Li L.** Obesity-related glomerulopathy: insights from gene expression profiles of the glomeruli derived from renal biopsy samples. *Endocrinology* 147: 44–50, 2006. doi:10.1210/en.2005-0641.
166. **Fowler SM**, **Kon V**, **Ma L**, **Richards WO**, **Fogo AB**, **Hunley TE.** Obesity-related focal and segmental glomerulosclerosis: normalization of proteinuria in an adolescent after bariatric surgery. *Pediatr Nephrol* 24: 851–855, 2009. doi:10.1007/s00467-008-1024-6.
167. **Shen WW**, **Chen HM**, **Chen H**, **Xu F**, **Li LS**, **Liu ZH.** Obesity-related glomerulopathy: body mass index and proteinuria. *Clin J Am Soc Nephrol* 5: 1401–1409, 2010. doi:10.2215/CJN.01370210.
168. **Redon J**, **Lurbe E.** The kidney in obesity. *Curr Hypertens Rep* 17: 555, 2015. doi:10.1007/s11906-015-0555-z.
169. **Jockers R**, **Delagrange P**, **Dubocovich ML**, **Markus RP**, **Renault N**, **Tosini G**, **Cecon E**, **Zlotos DP.** Update on melatonin receptors: IUPHAR Review 20. *Br J Pharmacol* 173: 2702–2725, 2016. doi:10.1111/bph.13536.
170. **Bhattacharya S**, **Patel KK**, **Dehari D**, **Agrawal AK**, **Singh S.** Melatonin and its ubiquitous anticancer effects. *Mol Cell Biochem* 462: 133–155, 2019. doi:10.1007/s11010-019-03617-5.
171. **Karamitri A**, **Jockers R.** Melatonin in type 2 diabetes mellitus and obesity. *Nat Rev Endocrinol* 15: 105–125, 2019. doi:10.1038/s41574-018-0130-1.
172. **Yawoot N**, **Govitrapong P**, **Tocharus C**, **Tocharus J.** Ischemic stroke, obesity, and the anti-inflammatory role of melatonin. *Biofactors* 47: 41–58, 2021. doi:10.1002/biof.1690.
173. **de Souza CAP**, **Gallo CC**, **de Camargo LS**, **de Carvalho PVV**, **Olesçuck IF**, **Macedo F**, **da Cunha FM**, **Cipolla-Neto J**, **do Amaral FG.** Melatonin multiple effects on brown adipose tissue molecular machinery. *J Pineal Res* 66: e12549, 2019. doi:10.1111/jpi.12549.
174. **Yin J**, **Li Y**, **Han H**, **Chen S**, **Gao J**, **Liu G**, **Wu X**, **Deng J**, **Yu Q**, **Huang X**, **Fang R**, **Li T**, **Reiter RJ**, **Zhang D**, **Zhu C**, **Zhu G**, **Ren W**, **Yin Y.** Melatonin reprogramming of gut microbiota improves lipid dysmetabolism in high-fat diet-fed mice. *J Pineal Res* 65: e12524, 2018. doi:10.1111/jpi.12524.
175. **Xu P**, **Wang J**, **Hong F**, **Wang S**, **Jin X**, **Xue T**, **Jia L**, **Zhai Y.** Melatonin prevents obesity through modulation of gut microbiota in mice. *J Pineal Res* 62: e12399, 2017. doi:10.1111/jpi.12399.
176. **Tofte N**, **Suvitaival T**, **Trost K**, **Mattila IM**, **Theilade S**, **Winther SA**, **Ahluwalia TS**, **Frimodt-Møller M**, **Legido-Quigley C**, **Rossing P.** Metabolomic assessment reveals alteration in polyols and branched chain amino acids associated with present and future renal impairment in a discovery cohort of 637 persons with type 1 diabetes. *Front Endocrinol (Lausanne)* 10: 818, 2019. doi:10.3389/fendo.2019.00818.
177. **Liu DJX**, **Stock E**, **Broeckx BJG**, **Daminet S**, **Meyer E**, **Delanghe JR**, **Croubels S**, **Devreese M**, **Nguyen P**, **Bogaerts E**, **Hesta M**, **Vanderperren K.** Weight-gain induced changes in renal perfusion assessed by contrast-enhanced ultrasound precede increases in urinary protein excretion suggestive of glomerular and tubular injury and normalize after weight-loss in dogs. *PLoS One* 15: e0231662, 2020. doi:10.1371/journal.pone.0231662.
178. **Winiarska K**, **Focht D**, **Sierakowski B**, **Lewandowski K**, **Orlowska M**, **Usarek M.** NADPH oxidase inhibitor, apocynin, improves renal glutathione status in Zucker diabetic fatty rats: a comparison with melatonin. *Chem Biol Interact* 218: 12–19, 2014. doi:10.1016/j.cbi.2014.04.005.
179. **Liu Z**, **Gan L**, **Xu Y**, **Luo D**, **Ren Q**, **Wu S**, **Sun C.** Melatonin alleviates inflammasome-induced pyroptosis through inhibiting NF-κB/GSDMD

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signal in mice adipose tissue. *J Pineal Res* 63: e12414, 2017. doi:10.1111/jpi.12414.

180. **Díaz-López A**, **Becerra-Tomás N**, **Ruiz V**, **Toledo E**, **Babio N**, **Corella D et al.** Effect of an intensive weight-loss lifestyle intervention on kidney function: a randomized controlled trial. *Am J Nephrol* 52: 45–58, 2021. doi:10.1159/000513664.
181. **Straznicky NE**, **Grima MT**, **Lambert EA**, **Eikelis N**, **Dawood T**, **Lambert GW**, **Nestel PJ**, **Masuo K**, **Sari CI**, **Chopra R**, **Mariani JA**, **Schlaich MP.** Exercise augments weight loss induced improvement in renal function in obese metabolic syndrome individuals. *J Hypertens* 29: 553–564, 2011. doi:10.1097/HJH.0b013e3283418875.
182. **Meloni C**, **Tatangelo P**, **Cipriani S**, **Rossi V**, **Suraci C**, **Tozzo C**, **Rossini B**, **Cecilia A**, **Di Franco D**, **Straccialano E**, **Casciani CU.** Adequate protein dietary restriction in diabetic and nondiabetic patients with chronic renal failure. *J Ren Nutr* 14: 208–213, 2004.
183. **Morales E**, **Praga M.** The effect of weight loss in obesity and chronic kidney disease. *Curr Hypertens Rep* 14: 170–176, 2012. doi:10.1007/s11906-012-0247-x.
184. **Bruci A**, **Tuccinardi D**, **Tozzi R**, **Balena A**, **Santucci S**, **Frontani R**, **Mariani S**, **Basciani S**, **Spera G**, **Gnessi L**, **Lubrano C**, **Watanabe M.** Very low-calorie ketogenic diet: a safe and effective tool for weight loss in patients with obesity and mild kidney failure. *Nutrients* 12: 333, 2020. doi:10.3390/nu12020333.
185. **Moinuddin I**, **Leehey DJ.** A comparison of aerobic exercise and resistance training in patients with and without chronic kidney disease. *Adv Chronic Kidney Dis* 15: 83–96, 2008. doi:10.1053/j.ackd.2007.10.004.
186. **Ikizler TA**, **Robinson-Cohen C**, **Ellis C**, **Headley SAE**, **Tuttle K**, **Wood RJ**, **Evans EE**, **Milch CM**, **Moody KA**, **Germain M**, **Limkunakul C**, **Bian A**, **Stewart TG**, **Himmelfarb J.** Metabolic effects of diet and exercise in patients with moderate to severe CKD: a randomized clinical trial. *J Am Soc Nephrol* 29: 250–259, 2018. doi:10.1681/ASN.2017010020.
187. **Welbourn R**, **Hollyman M**, **Kinsman R**, **Dixon J**, **Liem R**, **Ottosson J**, **Ramos A**, **Våge V**, **Al-Sabah S**, **Brown W**, **Cohen R**, **Walton P**, **Himpens J.** Bariatric surgery worldwide: baseline demographic description and one-year outcomes from the Fourth IFSO Global Registry Report 2018. *Obes Surg* 29: 782–795, 2019. doi:10.1007/s11695-018-3593-1.
188. **Chang AR**, **Chen Y**, **Still C**, **Wood GC**, **Kirchner HL**, **Lewis M**, **Kramer H**, **Hartle JE**, **Carey D**, **Appel LJ**, **Grams ME.** Bariatric surgery is associated with improvement in kidney outcomes. *Kidney Int* 90: 164–171, 2016. doi:10.1016/j.kint.2016.02.039.
189. **Imam TH**, **Fischer H**, **Jing B**, **Burchette R**, **Henry S**, **DeRose SF**, **Coleman KJ.** Estimated GFR before and after bariatric surgery in CKD. *Am J Kidney Dis* 69: 380–388, 2017. doi:10.1053/j.ajkd.2016.09.020.
190. **Shulman A**, **Peltonen M**, **Sjöström CD**, **Andersson-Assarsson JC**, **Taube M**, **Sjöholm K**, **Le Roux CW**, **Carlsson LMS**, **Svensson PA.** Incidence of end-stage renal disease following bariatric surgery in the Swedish Obese Subjects Study. *Int J Obes (Lond)* 42: 964–973, 2018. doi:10.1038/s41366-018-0045-x.
191. **Friedman AN**, **Wang J**, **Wahed AS**, **Docherty NG**, **Fennern E**, **Pomp A**, **Purnell JQ**, **Le Roux CW**, **Wolfe B.** The association between kidney disease and diabetes remission in bariatric surgery patients with type 2 diabetes. *Am J Kidney Dis* 74: 761–770, 2019. doi:10.1053/j.ajkd.2019.05.013.
192. **Mann JFE**, **Fonseca VA**, **Poulter NR**, **Raz I**, **Idorn T**, **Rasmussen S**, **von Scholten BJ**, **Mosenzon O**; **LEADER Trial Investigators.** Safety of liraglutide in type 2 diabetes and chronic kidney disease. *Clin J Am Soc Nephrol* 15: 465–473, 2020. doi:10.2215/CJN.11881019.
193. **Mann JFE**, **Fonseca V**, **Mosenzon O**, **Raz I**, **Goldman B**, **Idorn T**, **von Scholten BJ**, **Poulter NR.** Effects of liraglutide versus placebo on cardiovascular events in patients with type 2 diabetes mellitus and chronic kidney disease. *Circulation* 138: 2908–2918, 2018. doi:10.1161/CIRCULATIONAHA.118.036418.
194. **Sloan LA.** Review of glucagon-like peptide-1 receptor agonists for the treatment of type 2 diabetes mellitus in patients with chronic kidney disease and their renal effects. *J Diabetes* 11: 938–948, 2019. doi:10.1111/1753-0407.12969.
195. **Sarafidis P**, **Ferro CJ**, **Morales E**, **Ortiz A**, **Malyszko J**, **Hojs R**, **Khazim K**, **Ekart R**, **Valdivielso J**, **Fouque D**, **London GM**, **Massy Z**, **Ruggenenti P**, **Porrini E**, **Wiecek A**, **Zoccali C**, **Mallamaci F**, **Hornum M.** SGLT-2 inhibitors and GLP-1 receptor agonists for nephroprotection and cardioprotection in patients with diabetes mellitus and chronic kidney disease. A consensus statement by the EURECA-m and the DIABESITY working groups of the ERA-EDTA. *Nephrol Dial Transplant* 34: 208–230, 2019 [Erratum in *Nephrol Dial Transplant* 35: 1452, 2020]. doi:10.1093/ndt/gfy407.
196. **Cook SA**, **MacLaughlin H**, **Macdougall IC.** A structured weight management programme can achieve improved functional ability and significant weight loss in obese patients with chronic kidney disease. *Nephrol Dial Transplant* 23: 263–268, 2008. doi:10.1093/ndt/gfm511.
197. **Tong PC**, **Lee ZS**, **Sea MM**, **Chow CC**, **Ko GT**, **Chan WB**, **So WY**, **Ma RC**, **Ozaki R**, **Woo J**, **Cockram CS**, **Chan JC.** The effect of orlistat-induced weight loss, without concomitant hypocaloric diet, on cardiovascular risk factors and insulin sensitivity in young obese Chinese subjects with or without type 2 diabetes. *Arch Intern Med* 162: 2428–2435, 2002. doi:10.1001/archinte.162.21.2428.
198. **Leoncini G**, **Giovanna L**, **Viazzi F**, **Francesca V**, **Pontremoli R**, **Roberto P.** RAAS inhibition and renal protection. *Curr Pharm Des* 18: 971–980, 2012 [Erratum in *Curr Pharm Des* 19: 3748, 2013]. doi:10.2174/138161212799436520.
199. **Wolf G**, **Ritz E.** Combination therapy with ACE inhibitors and angiotensin II receptor blockers to halt progression of chronic renal disease: pathophysiology and indications. *Kidney Int* 67: 799–812, 2005. doi:10.1111/j.1523-1755.2005.00145.x.
200. **Ohmura T**, **Tsunenari I**, **Seidler R**, **Chachin M**, **Hayashi T**, **Konomi A**, **Matsumaru T**, **Sumida T**, **Hayashi N**, **Horie Y.** Renoprotective effects of telmisartan on renal injury in obese Zucker rats. *Acta Diabetol* 49, *Suppl* 1: S15–24, 2012. doi:10.1007/s00592-007-0016-5.
201. **Li H**, **Li M**, **Liu P**, **Wang Y**, **Zhang H**, **Li H**, **Yang S**, **Song Y**, **Yin Y**, **Gao L**, **Cheng S**, **Cai J**, **Tian G.** Telmisartan ameliorates nephropathy in metabolic syndrome by reducing leptin release from perirenal adipose tissue. *Hypertension* 68: 478–490, 2016. doi:10.1161/HYPERTENSIONAHA.116.07008.
202. **Mann JF**, **Schmieder RE**, **McQueen M**, **Dyal L**, **Schumacher H**, **Pogue J**, **Wang X**, **Maggioni A**, **Budaj A**, **Chaithiraphan S**, **Dickstein K**, **Keltai M**, **Metsärinne K**, **Oto A**, **Parkhomenko A**, **Piegas LS**, **Svendsen TL**, **Teo KK**, **Yusuf S**; **ONTARGET investigators.** Renal outcomes with telmisartan, ramipril, or both, in people at high vascular risk (the ONTARGET study): a multicentre, randomised, double-blind, controlled trial. *Lancet* 372: 547–553, 2008. doi:10.1016/S0140-6736(08)61236-2.
203. **Lattanzio MR**, **Weir MR.** Have we fallen off target with concerns surrounding dual RAAS blockade? *Kidney Int* 78: 539–545, 2010. doi:10.1038/ki.2010.225.
204. **Tylicki L**, **Rutkowski P**, **Renke M**, **Larczyński W**, **Aleksandrowicz E**, **Lysiak-Szydlowska W**, **Rutkowski B.** Triple pharmacological blockade of the renin-angiotensin-aldosterone system in nondiabetic CKD: an open-label crossover randomized controlled trial. *Am J Kidney Dis* 52: 486–493, 2008. doi:10.1053/j.ajkd.2008.02.297.
205. **Feng Y**, **Huang R**, **Kavanagh J**, **Li L**, **Zeng X**, **Li Y**, **Fu P.** Efficacy and safety of dual blockade of the renin-angiotensin-aldosterone system in diabetic kidney disease: a meta-analysis. *Am J Cardiovasc Drugs* 19: 259–286, 2019. doi:10.1007/s40256-018-00321-5.
206. **Abenavoli L**, **Scarpellini E**, **Colica C**, **Boccuto L**, **Salehi B**, **Sharifi-Rad J**, **Aiello V**, **Romano B**, **De Lorenzo A**, **Izzo AA**, **Capasso R.** Gut microbiota and obesity: a role for probiotics. *Nutrients* 11: 2690, 2019. doi:10.3390/nu11112690.
207. **Wang JW**, **Kuo CH**, **Kuo FC**, **Wang YK**, **Hsu WH**, **Yu FJ**, **Hu HM**, **Hsu PI**, **Wang JY**, **Wu DC.** Fecal microbiota transplantation: review and update. *J Formos Med Assoc* 118, *Suppl* 1: S23–S31, 2019. doi:10.1016/j.jfma.2018.08.011.
208. **Bäckhed F**, **Ding H**, **Wang T**, **Hooper LV**, **Koh GY**, **Nagy A**, **Semenkovich CF**, **Gordon JI.** The gut microbiota as an environmental factor that regulates fat storage. *Proc Natl Acad Sci USA* 101: 15718–15723, 2004. doi:10.1073/pnas.0407076101.
209. **Allegretti JR**, **Kassam Z**, **Mullish BH**, **Chiang A**, **Carrellas M**, **Hurtado J**, **Marchesi JR**, **McDonald JAK**, **Pechlivanis A**, **Barker GF**, **Miguéns Blanco J**, **Garcia-Perez I**, **Wong WF**, **Gerardin Y**, **Silverstein M**, **Kennedy K**, **Thompson C.** Effects of fecal microbiota transplantation with oral capsules in obese patients. *Clin Gastroenterol Hepatol* 18: 855–863.e2, 2020. doi:10.1016/j.cgh.2019.07.006.
210. **Vrieze A**, **Van Nood E**, **Holleman F**, **Salojärvi J**, **Kootte RS**, **Bartelsman JF**, **Dallinga-Thie GM**, **Ackermans MT**, **Serlie MJ**, **Oozeer R**, **Derrien M**, **Druesne A**, **Van Hylckama Vlieg JE**, **Bloks**

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VW, **Groen AK**, **Heilig HG**, **Zoetendal EG**, **Stroes ES**, **de Vos WM**, **Hoekstra JB**, **Nieuwdorp M.** Transfer of intestinal microbiota from lean donors increases insulin sensitivity in individuals with metabolic syndrome. *Gastroenterology* 143: 913–916.e7, 2012 [Erratum in *Gastroenterology* 144: 250, 2013]. doi:10.1053/j.gastro.2012.06.031.

211. **Holscher HD.** Dietary fiber and prebiotics and the gastrointestinal microbiota. *Gut Microbes* 8: 172–184, 2017. doi:10.1080/19490976.2017.1290756.

212. **Rossi M**, **Johnson DW**, **Morrison M**, **Pascoe EM**, **Coombes JS**, **Forbes JM**, **Szeto CC**, **McWhinney BC**, **Ungerer JP**, **Campbell KL.** Synbiotics easing renal failure by improving gut microbiology (SYNERGY): a randomized trial. *Clin J Am Soc Nephrol* 11: 223–231, 2016. doi:10.2215/CJN.05240515.

213. **McFarlane C**, **Krishnasamy R**, **Stanton T**, **Savill E**, **Snelson M**, **Mihala G**, **Kelly JT**, **Morrison M**, **Johnson DW**, **Campbell KL.** Synbiotics easing renal failure by improving gut microbiology II (SYNERGY II): a feasibility randomized controlled trial. *Nutrients* 13: 4481, 2021. doi:10.3390/nu13124481.

214. **Nguyen TTU**, **Kim HW**, **Kim W.** Effects of probiotics, prebiotics, and synbiotics on uremic toxins, inflammation, and oxidative stress in hemodialysis patients: a systematic review and meta-analysis of randomized controlled trials. *JCM* 10: 4456, 2021. doi:10.3390/jcm10194456.

215. **Guan Q**, **Wang Z**, **Cao J**, **Dong Y**, **Chen Y.** Mechanisms of melatonin in obesity: a review. *Int J Mol Sci* 23: 218, 2021. doi:10.3390/ijms23010218.

217. **Liakopoulos V**, **Franzén S**, **Svensson AM**, **Sattar N**, **Miftaraj M**, **Björck S**, **Ottosson J**, **Näslund I**, **Gudbjörnsdottir S**, **Eliasson B.** Renal and cardiovascular outcomes after weight loss from gastric bypass surgery in type 2 diabetes: cardiorenal risk reductions exceed atherosclerotic benefits. *Diabetes care* 43: 1276–1284, 2020. doi:10.2337/dc19-1703.

218. **Nehus EJ**, **Khoury JC**, **Inge TH**, **Xiao N**, **Jenkins TM**, **Moxey-Mims MM**, **Mitsnefes MM.** Kidney outcomes three years after bariatric surgery in severely obese adolescents. *Kidney Int* 91: 451–458, 2017. doi:10.1016/j.kint.2016.09.031.

219. **Solini A**, **Seghieri M**, **Santini E**, **Giannini L**, **Biancalana E**, **Taddei S**, **Volterrani D**, **Bruno RM.** Renal resistive index predicts post-bariatric surgery renal outcome in nondiabetic individuals with severe obesity. *Obesity (Silver Spring)* 27: 68–74, 2019. doi:10.1002/oby.22355.

220. **Neff KJ**, **Baud G**, **Raverdy V**, **Caiazzo R**, **Verkindt H**, **Noel C**, **Le Roux CW**, **Pattou F.** Renal function and remission of hypertension after bariatric surgery: a 5-year prospective cohort study. *Obes Surg* 27: 613–619, 2017. doi:10.1007/s11695-016-2333-7.

221. **Mann JFE**, **Ørsted DD**, **Brown-Frandsen K**, **Marso SP**, **Poulter NR**, **Rasmussen S**, **Tornøe K**, **Zinman B**, **Buse JB**; **LEADER Steering Committee and Investigators.** Liraglutide and renal outcomes in type 2 diabetes. *N Engl J Med* 377: 839–848, 2017. doi:10.1056/NEJMoa1616011.

222. **Tonneijck L**, **Smits MM**, **Muskiet MH**, **Hoekstra T**, **Kramer MH**, **Danser AH**, **Ter Wee PM**, **Diamant M**, **Joles JA**, **van Raalte DH.** Renal effects of DPP-4 inhibitor sitagliptin or GLP-1 receptor agonist liraglutide in overweight patients with type 2 diabetes: a 12-week, randomized, double-blind, placebo-controlled trial. *Diabetes care* 39: 2042–2050, 2016 [Erratum in *Diabetes Care* 42: 494, 2019]. doi:10.2337/dc16-1371.

223. **Stenlöf K**, **Rössner S**, **Vercruysse F**, **Kumar A**, **Fitchet M**, **Sjöström L**; **OBDM-003 Study Group.** Topiramate in the treatment of obese subjects with drug-naive type 2 diabetes. *Diabetes Obes Metab* 9: 360–368, 2007. doi:10.1111/j.1463-1326.2006.00618.x.

224. **Kang C**, **Qiao Q**, **Tong Q**, **Bai Q**, **Huang C**, **Fan R**, **Wang H**, **Kaliannan K**, **Wang J**, **Xu J.** Effects of exenatide on urinary albumin in overweight/obese patients with T2DM: a randomized clinical trial. *Sci Rep* 11: 20062, 2021. doi:10.1038/s41598-021-99527-y.

225. **Morales E**, **Gutiérrez E**, **Caro J**, **Sevillano A**, **Rojas-Rivera J**, **Praga M.** Beneficial long-term effect of aldosterone antagonist added to a traditional blockade of the renin-angiotensin-aldosterone system among patients with obesity and proteinuria. *Nefrologia* 35: 554–561, 2015. doi:10.1016/j.nefro.2015.09.008.

226. **Guney I**, **Selcuk NY**, **Altintepe L**, **Atalay H**, **Başarali MK**, **Büyükbaş S.** Antifibrotic effects of aldosterone receptor blocker (spironolactone) in patients with chronic kidney disease. *Ren Fail* 31: 779–784, 2009. doi:10.3109/08860220903150312.

Downloaded from journals.physiology.org/journal/ajpendo at Univ of Chicago (128.135.204.136) on March 8, 2026.